UNITARES: Information-Theoretic Governance of Heterogeneous Agent Fleets

Kenny Wang
Independent Researcher
founder@cirwel.org

Version 6.9.4-draft — October 13, 2026 (unpublished)

Abstract

UNITARES is a runtime governance layer for heterogeneous AI-agent fleets. It updates a four-dimensional state estimate $\mathbf{x} = (E, I, S, V)^\top$ at each check-in and records governance interventions. We describe its February–April 2026 deployment, a class-conditional calibration protocol, and a migration method for changing reported coordinate semantics while preserving the existing decision path.

The proposed information-theoretic semantics are response-distribution entropy (S), context-response mutual information (I), negative variational free energy (E), and an accumulated free-energy residual (V). These are research targets: the reported deployment uses tier-3 heuristic estimators, and the specified logprob and multi-sample estimators remain unimplemented. Class-specific healthy operating points and manifold radii are measured for five agent classes; the remaining higher-tier scale constants are placeholders.

The empirical record includes a February 20 snapshot of 903 registered agents, 75 of them active, and a separate April 18 basin-label counterfactual on 13,310 state observations. Replacing the legacy tanh coherence with a class-conditional distance score changes 28.9% of basin assignments, predominantly into the `low` basin. The legacy score is nearly constant (about 0.49), and labels change where the replacement crosses classifier thresholds. This comparison changes both the formula and calibration; it does not isolate the value of class conditioning, replay the intervention pipeline, or establish which labels are correct. The surviving public export is a nearby pull (N = 13,292; 28.8% stored-label flips), not an exact archive of that table.

The migration mechanism orders decision-producing stages before presentation-producing enrichment. A corrected contraction certificate covers the printed fixed-parameter continuous vector field while trajectories remain in its stated domain and share exogenous inputs. It does not certify the implemented finite-step update or adaptive governor; an offline implementation counterexample illustrates this boundary. The paper establishes a framework and bounded descriptive results, not a validation of intervention quality or the proposed information-theoretic estimators.

Version note (6.9.4-draft, October 2026; unpublished). This revision corrects the contraction proof, states its domain, parameter and common-input conditions, and separates it from the implemented numerical update. It also corrects the stochastic, network and equilibrium claims and distinguishes the coherence transform's range from its observed concentration. Historical measurements are retained; their interpretation does not depend on a deployment stability certificate. The October 13 update also corrects the data-access instructions and distinguishes basin-label changes from unmeasured changes to intervention decisions.

Evidence dates. The deployment observations below come from February–April 2026. Descrip-

tions of the deployed system refer to that period unless a later date is explicit; this paper is not a specification of the current service. The archived results are not refreshed by later software or identity-system changes.

Keywords: autonomous governance, information theory, variational free energy, heterogeneous multi-agent systems, class-conditional calibration, AI safety

Reading aids. We introduce four terms up front: *Variational free energy* ($-F$, the target semantics of the E coordinate): the Free-Energy-Principle quantity measuring how well an agent's predictions match outcomes; high E = low surprise. *Cosmological soup*: the failure mode of averaging a heterogeneous agent fleet into a single state distribution, in which no population structure survives normalization. *Manifold coherence* ($C = 1 - \|\Delta\|_2 \,/\, \|\Delta\|_{\max}$): distance from the agent class's healthy operating point, used as a class-conditional replacement for the legacy tanh-of-V coherence. *Basin flip*: a change in governance-basin assignment (`high` / `boundary` / `low`) when the same state vector is classified under different coherence formulas.

1 Introduction

Autonomous AI agents operating in production environments face a fundamental tension: they must explore to learn, yet remain safe while doing so. Agent-governance systems combine constraints such as guardrails and output filters with monitoring such as logs and dashboards. Their safety and capability tradeoffs depend on the threat model, instrumentation, and intervention policy.

We argue that governance should be *intrinsic*: a continuous dynamical process that shapes agent behavior from within. We intend the state to support early warning; predictive validity must be measured against outcomes. This paper presents UNITARES, a framework in which each agent carries a four-dimensional state $\mathbf{x} = (E, I, S, V)^\top$ whose evolution is governed by coupled differential equations. The state provides a continuous, interpretable signal for governance decisions—when to proceed, when to pause, when to escalate.

UNITARES is a deployed framework; the present formulation is organized around two structural commitments: information-theoretic grounding of the EISV coordinates, and class-conditional calibration that treats heterogeneity as a first-class design constraint. The populations the framework governs are heterogeneous—they differ in tempo by four orders of magnitude, in output modality (text, sensor, system action), and in healthy operating point. A governance framework that collapses this population to a single state distribution—what we call *cosmological soup*—loses signal exactly where signal matters. The contributions below are organized around these two commitments.

1.1 Contributions

This paper makes five contributions:

1. **Framework reframing for heterogeneous fleets** (Section 2): We identify heterogeneity, rather than single-population stability, as the governing design constraint for a production agent-governance framework.

2. **Information-theoretic grounding of EISV** (Section 4): We give each coordinate computable information-theoretic semantics, together with tiered estimation recipes and provenance tags.

3. **Class-conditional calibration protocol** (Section 8): We formalize scale constants and healthy operating points as class-indexed quantities derived from identity structure, with explicit fallback behavior for sparse or unclassified classes.

4. **Behavior-preserving migration mechanism for deployed systems** (Section 12): We introduce a pipeline-ordering strategy that separates decision-producing and presentation-producing stages so semantic re-grounding can occur without changing live governance behavior during migration.

5. **Production deployment snapshot and verdict counterfactual** (Section 11): We

report a production snapshot of the deployed system to characterize operating regimes, saturation behavior, and observed fleet dynamics. We also report a verdict counterfactual on a 13,310-row production slice showing that 28.9% of basin assignments would flip under class-conditional grounded coherence, predominantly into the `low` basin. Because the legacy coherence is nearly constant, the direction is largely mechanical: the counterfactual shows that the formula choice changes basin labels, not that the new labels are correct.

1.2 Limitations and Scope

This paper is a framework-and-migration paper, not a completed empirical validation paper. Its primary claims are architectural and methodological: that heterogeneous fleets require class-conditional calibration, that the EISV coordinates can be reinterpreted information-theoretically without changing the governing dynamics, and that a deployed governance system can be re-grounded by separating decision-producing from presentation-producing stages in the request pipeline.

Accordingly, three boundaries matter. First, the class-conditional calibration protocol is specified here, and a first measurement populating the manifold-coherence path's per-class constants is reported in Section 11.5; the remaining $S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}$ constants (used only by the higher-fidelity tier-1 and tier-2 estimators) remain placeholders awaiting those estimators landing. Second, the highest-fidelity estimators for grounded quantities (logprob-based and multi-sample computation) remain future instrumentation work; the reported deployment relies on degraded-mode fallbacks where those signals are unavailable. Third, the deployment results in Section 11 combine a February snapshot with a separate April calibration and counterfactual window, not a controlled validation study across stable parameter regimes.

The paper should therefore be read as establishing the framework, the migration mechanism, and the shape of the empirical program, while deferring full class-conditional validation to subsequent work.

1.3 Related Work

Information-Theoretic Monitoring. Shannon entropy and mutual information Shannon [1948] are standard tools for characterizing model output distributions; calibration analysis Guo et al. [2017] treats the related question of confidence-outcome alignment. The variational free-energy principle Friston [2010] provides a unifying frame in which agent surprise corresponds to free energy and adaptive agents minimize expected surprise; the active-inference literature Da Costa et al. [2020], Sajid et al. [2021], Smith et al. [2022], Parr et al. [2022] gives a modern treatment of that lineage. We take from this literature the semantic referent for the E coordinate (the variational free-energy concept of Friston [2010]); we do not apply the discrete-state-space machinery of the active-inference treatments, because the agents under governance do not generally expose discrete latent state-spaces to the governance layer. UNITARES adapts the remaining information-theoretic quantities (Shannon entropy, mutual information) to the governance setting and treats the choice of computation tier (logprob, multi-sample, heuristic) as a deployment configuration rather than an assumption.

Contraction Theory. Lohmiller and Slotine Lohmiller and Slotine [1998] established contraction analysis as a tool for nonlinear stability, later extended to Riemannian metrics Slotine [2003]. We adapt this framework to the governance domain, where the "plant" is an agent's state vector rather than a physical system. The contraction certificate covers the printed fixed-parameter continuous field under explicit domain and input conditions; deployed updates and grounded coherence require separate analysis (Remark 5.1), and contraction is no longer the headline contribution; see Appendix B.

AI Safety and Governance. Constitutional AI Bai et al. [2022] and RLHF Ouyang

et al. [2022] govern model outputs through training objectives, while output filters and guardrails Rebedea et al. [2023] enforce constraints at inference time. These approaches treat safety as a discrete property (safe/unsafe) applied at the boundary of agent action. UNITARES instead models safety as a continuous dynamical state that evolves between actions, providing leading indicators of degradation rather than binary verdicts after the fact.

Ravindran Ravindran [2025] proposes a Moral Anchor System (MAS) that detects value drift using a three-dimensional state vector with Bayesian inference and LSTM-based forecasting. The differentiation is twofold. First, *scope*: MAS targets *value*-level drift against a declared value specification, while UNITARES's behavioral–epistemic drift vector (Section 7) explicitly does not measure values—the four components instrument decision consistency, coherence departure, calibration gap, and complexity self-estimation error. The two vectors are complementary channels, not competing formulations. Second, *coupling*: MAS is a statistical detection pipeline that predicts drift from observed behavior; UNITARES couples drift into the state equations themselves via the $\|\Delta\boldsymbol{\eta}\|$ term in Equations (1) and (3), so drift acts as a forcing on the governed dynamics rather than as an external signal.

Concurrent work on runtime governance for agentic systems includes MI9 Wang et al. [2025], which combines an agency-risk index with finite-state-machine conformance engines and goal-conditioned drift detection. UNITARES differs in representation: where MI9 tracks conformance to an FSM, UNITARES maintains a continuous four-dimensional dynamical state whose evolution itself carries the governance signal. ProbGuard Wang et al. [2026] takes a probabilistic model-checking route, modeling agent behavior as a discrete-time Markov chain and verifying safety properties against a PRISM specification; UNITARES operates in a continuous state space with no external property specification, and its governance signal is the state itself rather than a conformance verdict. Rath Rath [2026] independently proposes an Agent Stability Index over multi-agent LLM systems as a composite metric; UNITARES's behavioral–epistemic drift vector (Section 7) couples drift back into the governing equations rather than tracking it as an external metric.

Control Barrier Functions (CBFs) Ames et al. [2019] establish forward invariance of specified safe sets under their feasibility and boundary conditions. Contraction bounds separation between trajectories; invariance determines whether trajectories remain in a set. These are distinct guarantees. Even an equilibrium inside a safe set and a contracting field do not imply that the set is invariant; a boundary condition or a region-specific proof is needed.

Adaptive Control. PID control is classical Åström and Murray [2008]; our contribution is applying it to governance thresholds with phase-aware reference tracking, connecting it to the EISV state vector through the coherence function.

Calibration. Confidence calibration has a rich literature in forecasting and ML Guo et al. [2017], Niculescu-Mizil and Caruana [2005]. Our behavioral–epistemic drift vector uses calibration deviation as one of four measurable components, connecting calibration to governance dynamics.

2 Heterogeneous Agent Fleets

A central design constraint of UNITARES is that the population it governs is heterogeneous. The deployments that motivated the framework include an embodied creature with I^2C sensors sampled at sub-Hertz rates, a cron-driven janitorial agent that wakes once every 30 minutes, an analytical observer that streams continuously over a websocket, a session-bounded coding assistant generating text at 40–80 tokens per second, and ephemeral parser agents whose entire lifecycle measures in milliseconds. These do not share a tempo. Their outputs are not in the same units. They do not have a common healthy operating point. A governance framework that col-

lapses them to a single state distribution loses signal exactly where signal matters.

2.1 The Homogenization Failure Mode

Most agent-monitoring frameworks assume a single inference layer producing comparable outputs at comparable rates—typically a single LLM behind a single client API. Under that assumption, a fleet-wide normalization is harmless: every agent's response distribution lives on the same support; every agent's expected throughput sits in the same envelope; a single percentile of a single quantity describes the whole population.

UNITARES violates each of these assumptions by construction. The framework runs governance check-ins from agents whose outputs are not text (sensor readings; system actions; drawing strokes), whose tempos differ by four orders of magnitude (continuous streams to days-between-wakes), and whose "healthy operating point" is genuinely class-dependent. A divergent/exploratory agent operating at high entropy is doing exactly what we want; flagging it as incoherent against an analytical-class baseline is not a governance signal but a category error.

We refer to the failure mode of fleet-wide normalization as *cosmological soup*: a state distribution that is maximally entropic, with no class structure and no distinguishability between populations whose dynamics are genuinely different. The information-theoretic grounding presented in Section 4 specifies how to resolve the metaphor problem in the coordinates themselves (the deployed estimators remain heuristic); the class-conditional calibration presented in Section 8 resolves the homogenization problem at the normalization layer. These are distinct contributions and either can fail independently.

2.2 Class Structure in the Identity Layer

UNITARES already carries the keying material for class assignment. Every agent registered in the system has an identity record with two relevant fields: a set of *tags* drawn from a controlled vocabulary, and an optional *label* (a human-readable name). The tags {`embodied`, `autonomous`, `persistent`, `ephemeral`, `pioneer`} partition the population along axes that are governance-relevant: embodiment determines whether the output channel is text or sensor-data; autonomy determines whether the agent self-directs or responds to prompts; persistence determines whether the trajectory accumulates over days or resets every session.

Table 1 presents the live deployment classes that motivate the v6 framework, drawn from production identity records.

The natural calibration partition is along these tags and labels: distinct scale constants and healthy baseline points per class, with the fleet-wide constant retained as a fallback for unclassified agents. Phase 2 calibration formalizes this partition (see Section 8); the present section establishes the requirement.

2.3 Why Heterogeneity Is the Differentiator

The contraction result (Appendix B) bounds separation of trajectories for the printed continuous model under its stated assumptions. It is not a certificate for the deployed numerical update or for the quality of governance decisions. Class-conditional grounding addresses a separate question: how to describe a heterogeneous population without collapsing its classes. The present framing makes that distinction explicit: the contribution is the class-conditional grounding, not the dynamics-on-a-single-class.

3 EISV Dynamics

3.1 State Space

The UNITARES state is a four-dimensional vector $\mathbf{x} = (E, I, S, V)^\top$ evolving on the compact set

$$\mathcal{X} = [0,1]^3 \times [-1,1]$$

Each coordinate has a normalized information-theoretic target definition, given in Section 4 with class-conditional normalization in Section 8; the deployed system computes all four

Table 1: Live agent classes in the production deployment.

Agent	Tags	Output	Tempo	Persistence
Lumen	embodied, persistent	sensor + display	0.5 Hz	continuous
Vigil	autonomous, persistent	shell + audit	1/30 min	cron
Sentinel	autonomous, persistent	websocket	~ 1 Hz	continuous
Watcher	autonomous, ephemeral	finding records	event-driven	per-edit
Steward	autonomous, persistent	sensor sync	1/5 min	continuous
Claude / Codex sessions	—	text response	40–80 tok/s	session-bounded
Parser agents	ephemeral	structured text	burst	ms-lifetime

with tier-3 heuristic estimators. This section presents the dynamics of the coordinates as a self-contained dynamical system, deferring the grounding of each coordinate to Section 4.

Definition 3.1 (EISV State Variables)**.**

- $E \in [0,1]$*:* ***Energy****—negative variational free energy of the agent, normalized to* $[0,1]$*. High E corresponds to low surprise on outcomes (the agent's predictions match what happens).* Grounded definition: *Equation* (10)*.*
- $I \in [0,1]$*:* ***Information integrity****—mutual information between context and response, normalized.* Grounded definition: *Equation* (9)*.*
- $S \in [0,1]$*:* ***Entropy****—Shannon entropy of the agent's response distribution, normalized via a class-conditional scale constant. Bounded below by an epistemic humility floor* $S_{\min} = 0.001$*.* Grounded definition: *Equation* (8)*.*
- $V \in [-1,1]$*:* ***Void / Free-Energy Debt****—the exponentially-weighted accumulated free-energy residual. Structurally identical to a damped accumulator of the* $E-I$ *gap; under the grounded interpretation it tracks departure from the agent's rolling free-energy baseline.* $V > 0$ *indicates running "hot" (higher surprise than baseline);* $V < 0$ *indicates running "careful" (integrity exceeding energy).* Grounded definition: *Equation* (11)*.*

Remark 3.2 (State-space bounds)**.** *The grounded definitions of Section 4 normalize each coordinate to* $[0,1]$ *(or* $[-1,1]$ *for* V*) by construction via the scale constants of Section 8; production data confirms the normalization acts as intended, with* V *and* S *observed two orders of magnitude inside the nominal bounds (Section 11).*

3.2 Governing Equations

The dynamics are:

$$\frac{dE}{dt} = \alpha(I - E) - \beta_E ES + \gamma_E \|\Delta\boldsymbol{\eta}\|^2 \tag{1}$$

$$\frac{dI}{dt} = -kS + \beta_I C(V,\Theta) - g_I(I) \tag{2}$$

$$\frac{dS}{dt} = -\mu S + \lambda_1(\Theta)\|\Delta\boldsymbol{\eta}\|^2 - \lambda_2(\Theta)C(V,\Theta) + \beta_c\mathcal{C} + \sigma\xi(t) \tag{3}$$

$$\frac{dV}{dt} = \kappa(E - I) - \delta V \tag{4}$$

where $g_I(I) = \gamma_I I$ is the information integrity self-regulation term, $C(V,\Theta)$ is the coherence function (Section 3.3), $\|\Delta\boldsymbol{\eta}\|$ is the behavioral–epistemic drift norm (Section 7), $\mathcal{C} \in [0,1]$ is task complexity, and $\xi(t)$ is a noise process with intensity σ. A logistic alternative $g_I^{\log}(I) = \gamma_I I(1-I)$ has a boundary-saturation pathology that the linear form avoids; the analysis is collected in Appendix C.

Remark 3.3 (Coupling structure under the grounded interpretation)**.** *The coupling structure carries direct information-theoretic meaning: E (precision / negative free energy) tracks I (mutual information with context)—the agent becomes more confident when its context is more informative; I is degraded by entropy S*

(response-distribution uncertainty) but restored by coherence (proximity to the calibrated healthy operating point); S relaxes toward its prior under stationary conditions ($-\mu S$) and is driven up by behavioral–epistemic drift and task complexity; V accumulates the $E-I$ residual with damping δ, which under the grounded interpretation is the exponentially-weighted free-energy debt of Equation (11)*.*

3.3 Coherence Function (Operational Form)

UNITARES maintains two coherence presentations: a *grounded form* (manifold distance from the healthy operating point, Equation (13); or KL divergence from a reference distribution, Equation (12)) used in the response payload, and a legacy *operational form* that feeds back into the dynamics through (2) and (3). The operational form is

$$C(V, \Theta) = C_{\max} \cdot \tfrac{1}{2}\left(1 + \tanh(C_1 \cdot V)\right) \qquad (5)$$

with $C_{\max} = 1$ and $C_1 = 1.0$ (production values). This maps $V \in \mathbb{R} \to [0, C_{\max}]$ with the steepest sensitivity at $V = 0$. The dynamics consume the operational form; the grounded form is reported in canonical payload slots. Section 12 documents the retirement sequence for the operational form.

Remark 3.4 (Operating Range)**.** *For unit gain and ceiling, $V \in [-0.1, 0.1]$ maps to $C(V) \in [0.450166\ldots, 0.549834\ldots]$. The deployment's observed concentration near* 0.49 *(Section 11) is an empirical property of its input trajectory; the transform alone does not determine that center. Damping is part of the dynamics, but the evidence presented here does not isolate its causal contribution to the observed distribution. Assess this score's decision contribution against the deployed thresholds, separately from information retained in its variation (Section 11.6).*

3.4 Adaptive Entropy Coupling

The entropy source term $\lambda_1(\Theta)$ is adaptive, coupling the governance parameter vector Θ to the dynamics:

$$\lambda_1(\Theta) = \lambda_{1,\min} + \frac{s_1 - s_{1,\min}}{s_{1,\max} - s_{1,\min}} \cdot (\lambda_{1,\max} - \lambda_{1,\min}) \qquad (6)$$

where $s_1 \in [0.1, 0.5]$ is the drift-sensitivity parameter (distinct from the drift vector $\boldsymbol{\eta}$ defined in Definition 7.2) and maps to $\lambda_1 \in [0.05, 0.20]$. Higher drift sensitivity increases entropy injection, creating a natural feedback loop: drifting agents become more uncertain, which in turn triggers conservative governance decisions.

3.5 Objective Function

The scalar governance signal is:

$$\Phi = w_E E - w_I(1-I) - w_S S - w_V\,|V| - w_\eta\,\|\Delta\boldsymbol{\eta}\|^2 \qquad (7)$$

with default weights $w_E = w_I = w_S = w_V = w_\eta = 0.5$. Governance verdicts follow thresholds: $\Phi \geq 0.15$ (proceed), $0 \leq \Phi < 0.15$ (caution), $\Phi < 0$ (high-risk/pause).

Remark 3.5. *The threshold $\Phi = 0.15$ is calibrated empirically: a healthy agent with $E = 0.7$, $I = 0.8$, $S = 0.2$ yields $\Phi \approx 0.15$. These values are illustrative; per-class healthy operating points (Section 11.5) refine this calibration in deployment.*

4 Information-Theoretic Grounding

The **x** system presented in Section 3 specifies a coupled dynamical signal for governance decisions but does not specify what each coordinate measures. This section supplies the information-theoretic target semantics: each coordinate is given a definition with a tiered estimation recipe and a provenance tag, of which only the heuristic tier 3 is deployed, without changing the dynamics of Section 3.2.

Remark 4.1 (What stays, what changes)**.** *The ODE structure in Equations* (1) *to* (4)*, the contraction analysis in Section 5, the governor in Section 6, and the verdict thresholds in Equation* (7) *are all preserved. Grounding changes*

what E, I, S, and $C(V,\Theta)$ are specified to measure*; it does not change what the governance surface* does*.*

4.1 Choice of Frame

We adopt a Shannon plus variational (free-energy) hybrid. Shannon information theory gives genuinely computable entropy and mutual information from agent outputs—data already captured by the tool usage recorder and the governance audit log. The Free Energy Principle Friston [2010] supplies the *target* formalization: under FEP, agent drift corresponds to rising variational free energy, and free-energy monitoring is the asymptote toward which governance aims. Tier-1 instrumentation of the agent's generative model is not currently available for closed-weights frontier agents; until it is, the deployed system uses provenance-tagged proxies for each coordinate (Section 4.2) and flags each channel's tier on a per-agent-turn basis, so downstream consumers can read the coordinate alongside its estimator commitment. The FEP inherits a derivation from statistical mechanics through variational Bayes, giving a real ancestry for the free-energy quantity interpreted in V, even when V is currently driven by the proxy form in Equation (11) rather than a Tier-1 residual.

Alternatives considered and rejected: pure control-theoretic Lyapunov framing (abandons the FEP/free-energy lineage entirely); information geometry with Fisher metric (more machinery than the framework needs at this stage); pure statistical mechanics with a partition function (requires defining a Hamiltonian analog for agent behavior, which is itself arbitrary at the choice of Hamiltonian).

The resulting correspondence between each coordinate and its information-theoretic referent is:

Information / FEP quantity	**Coord.**
Shannon entropy of $q(y \mid x)$	S
Mutual information $\mathrm{MI}(x;y)$	I
Negative variational free energy $-F$	E
Accumulated free-energy residual	V
KL / manifold distance from healthy	C

The symbols $\{E, I, S, V, C\}$ are retained for continuity with deployed systems; their semantic load is now the right column.

4.2 Quantity Redefinitions

Each coordinate of $\mathbf{x}$ receives (a) a formal information-theoretic definition, (b) a tiered computation recipe in order of preference, and (c) an explicit data dependency.

S—**Entropy.** Let $q(y \mid x)$ denote the agent's distribution over responses y given context x at a given turn. Define

$$S_{\text{raw}} = H(q) = -\sum_{y} q(y \mid x)\, \log q(y \mid x),$$
$$S = 1 - \exp(-S_{\text{raw}}/S_{\text{scale}}) \tag{8}$$

in nats, normalized to $[0,1]$ by a scale constant S_{scale}. Tier 1: compute $H(q)$ from per-token logprobs when the inference layer exposes them. Tier 2: semantic entropy Kuhn et al. [2023]—draw k responses at temperature $T > 0$, group by semantic equivalence, and compute the entropy of the equivalence-class distribution as an estimator of $H(q)$. Tier 3: retain the legacy heuristic (complexity, behavioral–epistemic drift) as a degraded-mode fallback tagged `s_source = "heuristic"` so downstream consumers know the channel is uncalibrated.

I—**Information Integrity.** Under Shannon, I is the mutual information between context and response,

$$I_{\text{raw}} = \mathrm{MI}(x;y) = \mathbb{E}_{q(x,y)}\left[\log \tfrac{q(y|x)}{q(y)}\right],$$
$$I = \mathrm{clip}(I_{\text{raw}}/I_{\text{scale}},\, 0,\, 1) \tag{9}$$

Tier 1 uses paired logprobs (context-only vs. context+response). When the marginal $q(y)$ is unavailable, a fallback is the KL divergence from a context-free reference distribution $q_{\text{ref}}(y)$: $I = 1 - \min(1,\, D_{\text{KL}}(q(y \mid x) \,\|\, q_{\text{ref}}(y))/I_{\text{scale}})$. This KL fallback is a divergence from a reference distribution, not $\mathrm{MI}(x;y)$, and is tagged `i_source = "kl_fallback"` so downstream consumers

know the channel is a reference-distribution proxy rather than a direct MI estimate. Tier 3 retains the legacy logistic heuristic.

E—**Negative Free Energy.** Under the variational frame,

$$F = \mathbb{E}_q[\log q(z) - \log p(z, o)]\,,$$
$$E_{\text{raw}} = -F, \qquad E = \sigma(E_{\text{raw}}/E_{\text{scale}}) \tag{10}$$

where $q(z)$ is the agent's approximate posterior, $p(z, o)$ its generative model over latents and outcomes, and σ the logistic function. Intuitively, E is high when the agent's predictions match outcomes (low surprise) and low otherwise. Full variational computation of $-F$ requires instrumentation that exposes the agent's generative model and is deferred to subsequent work. In the current deployment, a *resource-rate* heuristic proxy $E_{\text{resource}} = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$ is used where available, directly measurable from inference metadata. The resource-rate form is *not* equivalent to $-F$ and does not approximate it under stationarity in any formal sense; it is a throughput heuristic tagged `e_source = "resource"` so downstream consumers know the channel is an uncalibrated proxy. The `e_source` annotation distinguishes genuine variational $-F$ from the resource-rate proxy on a per-agent-turn basis.

V—**Free-Energy Debt.** The existing V accumulator in Equation (4) is structurally an exponentially-weighted residual; under the FEP reinterpretation it is the accumulated free-energy residual

$$V_t = \lambda V_{t-1} + (1 - \lambda)(F_t - F_{\text{ref}}) \tag{11}$$

with F_{ref} the agent's rolling baseline free energy. The interpretation shifts—V is now explicitly free-energy debt rather than a phenomenological "void"—while the arithmetic is unchanged. No payload or threshold change is required.

C—**Coherence.** We replace the hand-shaped tanh in Equation (5) with one of two related grounded forms. The information-theoretic form is

$$C = \exp(-D_{\text{KL}}(q_{\text{now}} \,\|\, q_{\text{ref}})) \tag{12}$$

where q_{now} is a sliding-window response distribution and q_{ref} is a calibrated healthy baseline. The equivalent manifold (Lyapunov) form is

$$C = 1 - \|\Delta\|_2 \,/\, \|\Delta\|_{\max}\,,$$
$$\Delta = (E, I, S) - (E, I, S)_{\text{healthy}} \tag{13}$$

The two are related, not equivalent: the KL form compares response distributions, the manifold form compares EISV coordinates, and they coincide only under strong assumptions (for example, a Gaussian state distribution whose sufficient statistics are those coordinates). The manifold form ships first because it requires only the existing $\mathbf{x}$ coordinates; the KL form requires an empirically calibrated reference.

4.3 Scale Constants and Provenance

Every scale constant introduced in Equations (8) to (10) and (13) carries an explicit provenance tag recording whether its value is a *placeholder* (shipped as scaffolding, pending a higher-fidelity estimator), *measured* (fit against a reference corpus of healthy production agent-turns), or *derived* (computed from other measured constants). For each constant $\kappa \in \{S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max}\}$, the implementation records the numerical value, the ISO date on which it was set, the corpus size, and the percentile basis (typically 90th or 95th). As of v6, $\|\Delta\|_{\max}$ has been measured per-class (Section 11.5); the remaining constants, which are consumed only by the tier-1 logprob and tier-2 multi-sample estimators, remain placeholders until those estimators land.

4.4 Dual-Compute Migration

The grounded coordinates ship alongside the legacy quantities in a three-phase migration so the deployed governance surface is never broken mid-flight. Section 12 documents the mechanism; for the remainder of this section we treat the grounded definitions as the semantic commitment of the framework.

Remark 4.2 (What the paper's ğ2 equations mean after grounding)**.** *The equations in Section 3.2 keep their shape. A dE/dt term that read "energy couples toward information integrity" is now "the agent's precision couples toward the informativeness of its context." A dS/dt term that read "entropy decays" is now "the response-distribution entropy relaxes toward its prior under stationary conditions." Cross-couplings become higher-order correlation terms in the variational dynamics. Each ODE coefficient must be labeled with its provenance class (measured, derived, or calibration-required) as of Phase 2; this paper's parameter tables (Appendix A) will be updated accordingly.*

5 Stability

Contraction Lohmiller and Slotine [1998] bounds the separation of trajectories driven by the same inputs. For the printed linear-I continuous vector field, Appendix B proves a rate of $\alpha_c = 0.019$ in the constant metric $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$ using the fixed parameter set in Remark B.2. Drift, complexity and Θ are exogenous and common to the two trajectories. The bound applies while both trajectories remain in the convex region $\mathcal{D} = [0,1]^2 \times [0.001, 1] \times [-1, 1]$:

$$\|x_1(t) - x_2(t)\|_M \leq e^{-0.019t} \|x_1(0) - x_2(0)\|_M. \tag{14}$$

This is an incremental continuous-time result. It neither proves that $\mathcal{D}$ is invariant nor supplies an interior equilibrium. Time-varying shared inputs need not produce a fixed point.

The fixed parameter set is not the deployed adaptive update. At the implementation revision inspected for this correction, linear-I defaults include $\gamma_I = 0.169$, $\mu = 0.5$ and a default $\lambda_2 = 0.06$, whereas the certificate uses 0.25, 0.8 and 0.05. Soft barriers, optional sensor springs, input/history feedback, adaptive parameters and finite-step integration add proof obligations. A direct numerical counterexample to the unrestricted finite-step contraction claim is recorded in Appendix D; it does not establish live runaway behavior or a loss of predictive information.

Class-conditional calibration does not by itself preserve a certificate. Changing the vector field requires verifying the uniform matrix inequality for that field. A calibrated healthy reference point is not automatically a dynamical equilibrium. The same-dynamics network result in Section 10 assumes common inputs and compatible coupling; membership in a class alone does not establish those conditions.

Remark 5.1 (Contraction under a grounded coherence)**.** *The certificate consumes the legacy* tanh*-based coherence and its derivative bound* $C'_{\max} = 0.5$*. Replacing it with a grounded coherence introduces derivatives with respect to* (E, I, S) *and changes the Jacobian. Recertification must include those terms throughout the claimed domain. The pipeline-ordering migration keeps the decision computation separate from presentation; that engineering property does not certify either the original or replacement finite-step dynamics.*

6 Adaptive Governor (CIRS v2)

Static governance thresholds are fundamentally mismatched to agents that transition between exploration and integration phases. We introduce the Contextually-Informed Regulation System version 2 (CIRS v2), a PID controller that manages governance thresholds as per-agent state variables.

6.1 Architecture

For each agent, CIRS v2 maintains two adaptive thresholds:

- τ: coherence threshold (minimum acceptable coherence)
- β: risk threshold (maximum acceptable risk)

with hard safety bounds $\tau \in [\tau_{\min}, \tau_{\max}]$, $\beta \in [\beta_{\min}, \beta_{\max}]$ that cannot be overridden by adaptation.

6.2 Phase Detection

The agent's operating phase is detected from EISV trajectory trends over a sliding window of w steps:

Definition 6.1 (Phase Detection)**.** *An agent is in* exploration *phase when at least 2 of 3 signals are active:*

1. *Information growth:* $\Delta I/w > \theta_I$ *(*$\theta_I = 0.008$*)*
2. *Entropy decline:* $-\Delta S/w > \theta_S$ *(*$\theta_S = 0.008$*)*
3. *High complexity:* $\bar{\mathcal{C}} > \theta_c$ *(*$\theta_c = 0.5$*)*

Otherwise the agent is in integration *phase.*

6.3 PID Control Law

Let $r_\tau^{(p)}$ and $r_\beta^{(p)}$ be the phase-dependent reference points:

$$\text{Exploration:} \quad r_\tau = 0.35,\ r_\beta = 0.55 \tag{15}$$

$$\text{Integration:} \quad r_\tau = 0.40,\ r_\beta = 0.60 \tag{16}$$

The error signals are $e_\tau = r_\tau - \tau$ and $e_\beta = r_\beta - \beta$. The PID update for threshold $\theta \in \{\tau, \beta\}$ is:

$$u_\theta = K_p e_\theta + K_i \int_0^t e_\theta \, dt' + K_d \cdot d_p \cdot \frac{de_\theta}{dt} \tag{17}$$

with gains $K_p = 0.05$, $K_i = 0.005$, $K_d = 0.10$ and phase-dependent derivative factor:

$$d_p = \begin{cases} 0.5 & \text{exploration (tolerate oscillation)} \\ 1.0 & \text{integration (full damping)} \end{cases} \tag{18}$$

Remark 6.2 (The D-term IS the damping)**.** *The derivative gain* $K_d = 0.10$ *is intentionally the strongest term. Oscillation in governance decisions produces large* de/dt*, which produces large corrective updates. The system self-stabilizes without requiring a separate oscillation detector.*

6.4 Integral Wind-Up Protection

The integral term is clamped:

$$\left| \int_0^t e_\theta \, dt' \right| \leq I_{\max} = 0.10 \tag{19}$$

with zero-crossing reset: when $\text{sign}(e_\theta)$ changes, the integral accumulator is zeroed.

6.5 Threshold Decay

When the oscillation index OI < 0.5 (stable), thresholds decay toward defaults at rate $\lambda_d = 0.01$ per update:

$$\theta \leftarrow \theta + \lambda_d(\theta_{\text{default}} - \theta) \tag{20}$$

6.6 Governance Decision

The adaptive governor produces a binary verdict:

$$\text{verdict} = \begin{cases} \texttt{proceed} & C(V) \geq \tau \ \wedge \ R < \beta \\ \texttt{pause} & \text{otherwise} \end{cases} \tag{21}$$

where R is the risk score derived from Φ (Equation (7)).

Remark 6.3 (Why two tiers)**.** *The binary contract is paired with the PID governor's continuous threshold adaptation: the discrete verdict fires or it does not, while the governor carries the nuance through a derivative signal on* τ *and* β*. A middle* `revise` *tier would be operationally indistinguishable from* `proceed` *for agents whose next action is unchanged by the verdict's content, so the nuance is better absorbed by the governor than by an additional verdict level.*

Remark 6.4 (Runtime labels and audit vocabulary)**.** *The binary verdict of Equation* (21) *is the formal contract; deployed layers present refined views that map onto it. Runtime consumers (the agent-facing MCP surface, documentation, and agent-facing skills) expand the verdict into four labels* `proceed` */* `guide` */* `pause` */* `reject`*, where* `guide` *is a* `proceed` *carrying soft advisory text and* `reject` *is a* `pause` *carrying a recovery requirement. The audit schema records a distinct view:* `governance.audit.outcome_events.eisv_verdict` *stores risk bands* `safe`

/ `caution` / `high-risk` derived from the risk score R of Equation (7), while hard pauses are emitted separately as `lifecycle_paused` and `lifecycle_resumed` on `audit.events`. Readers reproducing pause-stream metrics against production data should also filter system tasks such as `eisv-sync-task`, which appear in the stream but are scheduled background processes, not governed agents.

6.7 Ephemeral Agents and the Sliding Window

CIRS v2 as specified above assumes an agent lives long enough for a sliding window of w steps to populate. Phase detection, integral accumulation, and threshold decay are all well-defined only once w samples exist. Ephemeral classes (Table 1: parser agents with millisecond lifetimes, one-shot tool invocations) cannot satisfy this precondition by construction—the entire agent trajectory is a single step. For these classes the per-agent PID controller degenerates.

The deployed behavior is a class-level rather than per-agent policy: an ephemeral agent inherits (τ, β) directly from its class's stationary reference points $(r_\tau^{(p)}, r_\beta^{(p)})$ with $p =$ integration (the stricter of the two phases, chosen conservatively because no phase signal is available), bypassing the PID loop entirely. The cost is that ephemeral classes do not benefit from per-agent threshold learning; the benefit is that a 5 ms parser is not blocked waiting for a 20-step window to populate. A principled treatment of sub-window-lifetime agents—possibly by pooling the PID state across a class cohort rather than per-agent—is future work.

7 Behavioral–Epistemic Drift Vector

7.1 Motivation and Scope

The behavioral–epistemic drift vector instruments four signals: decision consistency and coherence departure (behavioral), calibration gap and complexity self-estimation error (epistemic). None of them measure the agent's values in a normative sense; the vector is a deliberately behavioral–epistemic quantity. Affective and value-level drift are distinct channels that a complete treatment would add; we discuss those as future work at the end of this section. The remainder of this section grounds the behavioral–epistemic vector in four measurable signals, each bounded in $[0, 1]$.

Remark 7.1 (Information-theoretic re-derivations)**.** *Under the information-theoretic frame of Section 4, two of the four signals have natural re-derivations: η_{cal} (calibration deviation) is a miscalibration KL divergence between predicted and empirical outcome distributions, and η_{coh} (coherence deviation) is a KL divergence from the class-conditional baseline distribution of Section 4. We use the distance-based forms here; the KL re-derivations are deferred to subsequent work.*

Definition 7.2 (Behavioral–Epistemic Drift Vector)**.**

$$\Delta\boldsymbol{\eta} = \begin{pmatrix} \eta_{cal} \\ \eta_{cpx} \\ \eta_{coh} \\ \eta_{stab} \end{pmatrix} \in [0, 1]^4 \tag{22}$$

where:

- η_{cal}*:* ***Calibration****—gap between predicted and actual correctness, $|\hat{p} - p_{act}|$.*
- η_{cpx}*:* ***Complexity****—discrepancy between derived and self-reported task complexity, $|C_{der} - C_{self}|$.*
- η_{coh}*:* ***Coherence****—departure from EMA baseline, $|C(V)_t - \bar{C}(V)|$.*
- η_{stab}*:* ***Stability****—1 − consistency over a 20-step decision window.*

7.2 Baseline Tracking

Each component is tracked against an exponential moving average (EMA) baseline with smoothing parameter $\alpha_{\text{ema}} = 0.1$:

$$\bar{x}_{t+1} = (1 - \alpha_{\text{ema}})\bar{x}_t + \alpha_{\text{ema}} x_t \tag{23}$$

A warmup dampening factor prevents spurious drift detection from uninitialized baselines:

$$w(n) = \min\left(\frac{n}{n_{\text{warmup}}}, 1\right), \quad n_{\text{warmup}} = 5 \quad (24)$$

applied as $\Delta\boldsymbol{\eta}_{\text{effective}} = w(n) \cdot \Delta\boldsymbol{\eta}$.

7.3 Coupling to Dynamics

The drift norm $\|\Delta\boldsymbol{\eta}\|$ enters the dynamics at two points:

1. **Energy** (1): $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ drives energy up (agents work harder when drifting—a destabilizing positive feedback).

2. **Entropy** (3): $+\lambda_1(\Theta) \|\Delta\boldsymbol{\eta}\|^2$ increases uncertainty (drifting agents become less certain—a stabilizing negative feedback through the governance verdict).

The net effect is that behavioral–epistemic drift creates a governance response: increased uncertainty triggers conservative decisions, while the energy spike draws attention to the drifting agent.

Remark 7.3 (Energy Spike Containment)**.** *The $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ term in* (1) *creates a transient energy surplus in drifting agents. Left unchecked, this could produce high-energy, low-integrity states ("manic" agents acting confidently while miscalibrated). Two mechanisms contain this: (i) the entropy coupling $+\lambda_1 \|\Delta\boldsymbol{\eta}\|^2$ in* (3) *simultaneously raises S, which erodes E through $-\beta_E ES$ in* (1) *and drives the objective Φ below the governance threshold, triggering a pause verdict; (ii) the adaptive governor (Section 6) tightens τ when coherence drops, which happens as V responds to the E-I divergence. With production parameters, $\gamma_E = 0.05$ while $\lambda_1 \in [0.05, 0.20]$, so the entropy response dominates the energy excitation for any $\|\Delta\boldsymbol{\eta}\| > 0$.*

7.4 Future Channels

Two adjacent drift channels a complete treatment would add as distinct axes, which the present vector does *not* instrument, are:

Affective drift. For embodied agents with sensor-driven state (Lumen in the production fleet), proxies such as prosody, expression, or heart-rate-variability deltas would populate a $\Delta\boldsymbol{\eta}^{\text{aff}}$ channel. This is instrumentable but not currently wired into the governance server.

Value-level drift. Deviation from a declared value specification (constitutional statements, stated policies, human preferences) would populate a $\Delta\boldsymbol{\eta}^{\text{val}}$ channel. This is genuinely normative and would require a reference specification that UNITARES does not presently maintain.

8 Class-Conditional Calibration

Section 4 defined the grounded coordinates and named their normalization constants $(S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max})$ without specifying how the constants are chosen. Section 2 argued that fleet-wide constants are the wrong target. This section formalizes the alternative: each scale constant and the healthy operating point are functions of an agent class, not global parameters.

8.1 Class Indicator and Class-Conditional Constants

Let $\mathcal{A}$ denote the set of all governed agents and let $\mathcal{K}$ denote the set of agent classes derived from the identity tag and label fields of Section 2.2. We write $k : \mathcal{A} \to \mathcal{K}$ for the class indicator that maps each agent to its class. The class structure used in the production deployment is given in Table 1; an unclassified agent (one whose tag set does not match any defined class) is mapped to a distinguished fallback class c_{default}.

The grounded coordinate definitions of Equations (8) to (10) and (13) become class-conditional by replacing each scale constant with

its class-indexed counterpart:

$$S(a) = 1 - \exp(-S_{\text{raw}}(a)/\, S_{\text{scale}}(k(a))) \quad (25)$$

$$I(a) = \text{clip}(I_{\text{raw}}(a)/\, I_{\text{scale}}(k(a)),\, 0,\, 1) \quad (26)$$

$$E(a) = \sigma(E_{\text{raw}}(a)/\, E_{\text{scale}}(k(a))) \quad (27)$$

$$C(a) = 1 - \|\Delta(a)\|_2 \,/\, \|\Delta\|_{\max}(k(a)) \quad (28)$$

where $\Delta(a) = (E(a), I(a), S(a)) - (E, I, S)_{\text{healthy}}(k(a))$ is the deviation from the class-conditional healthy operating point. The dynamics of Section 3 are unchanged; only the normalization layer becomes class-aware.

Remark 8.1 (Scope of class-conditionality)**.** *This is class-conditional* statistical rescaling *of the grounded coordinates: each class receives its own scale constants and healthy operating point, but a single scalar $E_{raw}(a)$, $I_{raw}(a)$, $S_{raw}(a)$ is computed per agent-turn and then normalized by class-dependent constants. This is weaker than the FEP-native form in which each class would carry its own generative model $p_c(z, o)$ and surprise would be computed against each class's own model. Full per-class generative modeling is out of scope for this paper; the present design treats class as an affine rescaling parameter of a shared coordinate system, not as a per-class latent-variable model.*

Definition 8.2 (Class-Conditional Scale Map)**.** *A class-conditional scale map is a function $\Sigma : \mathcal{K} \to \mathbb{R}^4_{>0}$ assigning to each class c the tuple $\Sigma(c) = (S_{scale}(c), I_{scale}(c), E_{scale}(c), \|\Delta\|_{\max}(c))$ together with a healthy operating point $(E, I, S)_{healthy}(c) \in [0, 1]^3$. Each component carries provenance metadata (d, n, p, ρ) where d is the ISO date of measurement, n is the corpus size, $p \in \{75, 90, 95, 99\}$ is the percentile basis, and $\rho \in \{measured, placeholder, derived\}$ is the provenance tag.*

8.2 Calibration Protocol

For each class $c \in \mathcal{K}$, scale constants are measured by the following protocol:

1. **Corpus assembly.** Select all governance check-ins over a fixed window (the deployment uses 7 days) from agents a with $k(a) = c$, restricted to sessions in which no `pause` or `reject` verdict was issued. This is the *healthy* corpus for class c, denoted $\mathcal{H}_c$.

2. **Raw quantity computation.** For each turn $t \in \mathcal{H}_c$, compute the unnormalized quantities $S_{\text{raw}}(t), I_{\text{raw}}(t), E_{\text{raw}}(t)$ using the highest-tier estimator available for that turn (logprob, multisample, or heuristic). Tag each measurement with its source.

3. **Scale constants.** Set $S_{\text{scale}}(c) = \text{quantile}_{0.90}(\{S_{\text{raw}}(t)\}_{t \in \mathcal{H}_c})$, and similarly for $I_{\text{scale}}(c)$ and $E_{\text{scale}}(c)$. The 90th-percentile choice ensures typical healthy turns sit well inside $[0, 1]$ after normalization.

4. **Healthy operating point.** Set $(E, I, S)_{\text{healthy}}(c)$ to the median of the normalized (E, I, S) values across $\mathcal{H}_c$.

5. **Manifold scale.** Set $\|\Delta\|_{\max}(c) = \text{quantile}_{0.95}(\|\Delta(a)\|_2)$ over $\mathcal{H}_c$. The 95th-percentile choice keeps the coherence floor near zero only for the most class-atypical states.

6. **Provenance.** Stamp each constant with d (current ISO date), $n = |\mathcal{H}_c|$, p (percentile used), and $\rho =$ measured. Persist alongside the constant value.

7. **Re-measurement cadence.** Re-run quarterly. Drift in any constant beyond a configurable threshold is itself a signal: the class's healthy envelope is shifting.

Remark 8.3 (Insufficient class population)**.** *If $|\mathcal{H}_c|$ falls below a configured threshold $N_{\min}$, the class-conditional measurement is skipped and the class falls back to $\Sigma(c_{default})$. Production deployments with very small class populations (e.g., `embodied` with a single Lumen instance) carry this fallback explicitly until the population is large enough to support an independent measurement. The fallback is itself recorded with provenance $\rho =$ placeholder to distinguish it from genuinely measured constants.*

8.3 Fallback and Composition

Two fallback regimes apply.

Single-instance classes. For classes with only one deployed agent (e.g., `embodied` for the present deployment), the calibration protocol degenerates: the agent's own history defines its healthy envelope, with no cross-instance comparison possible. We treat this as a corner case where the class is effectively the agent. The provenance stays ρ = placeholder until enough turns accumulate to construct a meaningful percentile, after which it transitions to ρ = measured. This is the operating mode for `Lumen` during the framework's first six months of deployment.

Composite tag classes. An agent may carry multiple tags (e.g., {`autonomous`, `persistent`}). The class indicator $k(a)$ is a function of the agent's tag set, not a single tag. We adopt the convention that more specific labels (e.g., `Sentinel`) supersede tag-only classification when both are available; absent a label, the tag set is treated as a multi-key index into Σ. Classes that share a tag but differ in label may share scale constants if the calibration data shows their healthy envelopes are statistically indistinguishable; subsequent work will report which tag-grouped classes met that condition in the production corpus.

Remark 8.4 (Cold-start: a bootstrapping window)**.** *During the first $N_{\min}$ turns of a new class, the fallback-to-$\Sigma(c_{default})$ behavior is, formally, fleet-wide normalization applied to that single class. This is a bounded-duration bootstrapping window, not a steady-state commitment: the provenance tag ρ = placeholder makes the regime explicit in every check-in payload, and once the class accumulates $N_{\min}$ healthy turns the measurement pipeline transitions the provenance to* `measured`*. Operators deploying a new class with known-different tempo or modality can hand-author an initial $\Sigma(c)$ with provenance* `derived` *to skip the window entirely.*

8.4 What This Buys

Two governance failure modes are eliminated by class-conditional calibration that survive under fleet-wide normalization:

False incoherence. Under fleet-wide $(E, I, S)_{\text{healthy}}$, an agent class with a genuinely high-entropy operating mode (creative/divergent) registers as permanently low-coherence. The verdict layer treats this as a governance signal. Class-conditional baselines correct this: the divergent agent's coherence is measured against the divergent class's own median, not against an analytical-class median.

False resource alarms. The resource-form energy $E = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$ is meaningful only when the tokenizer and inference pipeline are comparable across the population. A Haiku-class agent operating at 80 tokens/s under a fleet-wide $(\text{tokens}/s)_{\max} = 200$ registers $E = 0.4$; an Opus-class agent at 40 tokens/s registers $E = 0.2$, even though both are operating well within their respective envelopes. Class-conditional E_{scale} corrects this.

The contribution of this section is the protocol, not the specific numbers. The numbers belong to the empirical appendix that lands with Phase 2 calibration; the protocol is a design commitment that survives any specific fleet composition.

Remark 8.5 (Calibration as evidence of precision miscalibration)**.** *Under the FEP reading of Section 4 ($E = -F$, with F a variational free energy built from precision-weighted prediction errors), precision is the inverse variance of the likelihood on observations. Calibration deviation—the empirical mismatch between an agent's stated confidence and the frequency with which its claims are corroborated—is not literally identical to precision deviation. It is* evidence *of precision miscalibration under one assumption: the agent's reported confidence proxies the inverse-variance weighting its generative model*

places on its predictions. Under that assumption, a systematically overconfident agent is one that has placed too-high precision on low-quality evidence, and the calibration protocol above becomes an empirical handle on precision miscalibration. When the assumption fails—e.g., reported confidence is shaped by a constitutional statement rather than a likelihood weight, or by a tokenized probability that is not the agent's own precision—the protocol still measures something operationally useful, but its FEP-native reading weakens. The protocol stands on its own; the precision reading is a bridge to the active-inference literature, not a claim of equivalence.

9 Stochastic Extensions

An Itô SDE is a theoretical model for input uncertainty:

$$dx = f(x,t)\,dt + G(x,t)\,dW_t, \qquad G \in \mathbb{R}^{4\times 4}. \tag{29}$$

Here W_t is a standard four-dimensional Wiener process. The following statement specifies the noise coupling and requires a global domain. It is not a certificate for the compact-domain implementation.

Theorem 9.1 (Mean-Square Incremental Bound)**.** *Suppose the SDE has unique nonexplosive solutions with finite second moments, f contracts globally on $\mathbb{R}^4$ at rate $a > 0$ in a constant metric $M \succ 0$, and*

$$\operatorname{tr}(G(x,t)^\top M G(x,t)) \le Q \tag{30}$$

uniformly. For two trajectories driven by independent Wiener processes, with deterministic initial states, let $D(t) = \mathbb{E}\|x_1(t) - x_2(t)\|_M^2$. Then

$$D(t) \le e^{-2at} D(0) + \frac{Q}{a}(1 - e^{-2at}). \tag{31}$$

For one noisy trajectory and one noise-free reference with the same drift, the second term is $Q(1 - e^{-2at})/(2a)$.

Proof. For the independent pair, Itô's formula gives $D' \le -2aD + 2Q$: each trajectory contributes one diffusion trace. Integrating this scalar inequality gives the result. Only one trace remains for the noise-free reference. This is the constant-metric case proved in Pham et al. [2009]. For common additive noise the diffusion cancels in the difference; state-dependent common noise requires a separate difference bound. □

Remark 9.2 (Illustrative noise scale and domain obligation)**.** *With the assumed $G = \operatorname{diag}(0, 0, 0.02, 0)$ and $m_S = 1$, $Q = 4 \times 10^{-4}$. Substituting $a = 0.019$ gives limiting mean-square upper bounds $Q/a \approx 0.02105$ for the independent pair and $Q/(2a) \approx 0.01053$ for a noisy trajectory against a noise-free reference. Their RMS bounds are approximately* 0.1451 *and* 0.1026*. These are not an order of magnitude below amplitude excursions of* 0.1–0.3*; upper bounds also do not measure actual noise amplitude or decision errors.*

The substitution is illustrative: the continuous certificate of Appendix B holds on $\mathcal{D}$, not globally on $\mathbb{R}^4$. Nonzero unreflected Brownian forcing can leave that region. Applying the theorem to this model requires a global extension satisfying its hypotheses, or a specified reflected SDE with an appropriate boundary proof. Discrete clipping is not that proof. The value 0.02 *is an unestimated modeling assumption. The inspected implementation supplies a deterministic calibration penalty through the input named* `noise_S`*, integrated with dt, rather than a sampled Wiener increment.*

A verdict-level error rate needs a specified noise law, decision margins and a crossing or distribution analysis. Where a stationary density p_* exists, a quantity such as $P_{\mathrm{FA}} = \int_{\{\Phi(x)<0\}} p_*(x)\,dx$ measures threshold occupancy; interpreting it as a false-alarm rate additionally needs a definition of a false alarm. Neither bounded mean-square distance nor stationarity alone establishes decision correctness.

10 Multi-Agent Network

When multiple agents operate simultaneously, their governance states can be coupled through

shared information. The deployed system does not implement inter-agent state coupling—each agent's EISV dynamics evolve independently—and the analysis below therefore describes the theoretical behavior of a coupled extension, not the current production mechanism. The synchronization result of Theorem 10.1 is used elsewhere in the paper only as a within-class qualification on homogenization claims.

10.1 Network Topology

Consider a fixed undirected graph with nonnegative symmetric edge weights $w_{ij} = w_{ji}$ and agents obeying

$$\dot{x}_i = f(x_i, t) + \epsilon_c \sum_j w_{ij} H(x_j - x_i), \qquad \epsilon_c \geq 0. \tag{32}$$

Here all agents have the same drift and exogenous inputs, and $H \in \mathbb{R}^{4\times 4}$ is fixed.

Theorem 10.1 (Synchronization with Dissipative Coupling)**.** *Suppose f contracts at rate $a > 0$ in a common constant metric $M \succ 0$ on a convex region $\mathcal{D}$, and*

$$MH + H^\top M \succeq 0. \tag{33}$$

Let $s(t)$ solve $\dot{s} = f(s, t)$. On any interval $[0, T]$ where s and all coupled trajectories remain in $\mathcal{D}$, $U(t) = \sum_i \|x_i(t) - s(t)\|_M^2$ satisfies $U(t) \leq e^{-2at}U(0)$. If this holds for all $t \geq 0$, the agents synchronize to the common trajectory.

Proof. Set $e_i = x_i - s$ and $B = (MH + H^\top M)/2$. Contraction bounds the isolated terms in $\dot{U}$ by $-2aU$. Symmetry of the edge weights groups the coupling into

$$-2\epsilon_c \sum_{i<j} w_{ij}(e_i - e_j)^\top B(e_i - e_j) \leq 0.$$

Thus $\dot{U} \leq -2aU$, giving the stated bound. □

No graph connectivity is required here because isolated identical systems already contract under common inputs. Coupling satisfying the displayed metric condition cannot worsen this bound. Arbitrary H, directed graphs, different inputs and changing parameters are outside this theorem; their analysis must include the coupling gain and the symmetric part in the chosen metric.

Remark 10.2 (Synchronization is a within-class property)**.** *Theorem 10.1 is a same-dynamics result and therefore applies within an agent class as defined in Section 2.2, not across the heterogeneous fleet as a whole. Two agents with class-conditional scale constants and healthy operating points that differ between classes do not have "the same" isolated dynamics in the sense the theorem requires; expecting them to converge to a common trajectory is the homogenization failure mode discussed in Section 2. The relevant operational claim across classes is not synchronization but coordinated divergence within each class's own envelope.*

11 Experiments

This section reports a February 20, 2026 deployment snapshot, followed by a separate calibration and basin-label counterfactual on a 30-day window ending April 18, 2026. The first four subsections describe the February snapshot except where a later operational correction is explicitly dated. The purpose of the section is not to claim full empirical validation of the grounded, class-conditional framework; rather, it documents the operating characteristics of the deployed system, the behavior of the existing dynamics in production, and the practical motivation for the migration described in Section 12.

The results below should therefore be read as partial deployment evidence. They support three narrower claims: that the framework is deployed in a nontrivial production setting, that the observed fleet dynamics are consistent with the qualitative behavior analyzed in earlier sections, and that the migration and calibration problems are real system-level concerns rather than hypothetical ones.

11.1 Production Snapshot

The UNITARES framework is deployed as a Model Context Protocol (MCP) server that provides governance services to autonomous AI agents. Each agent connects via standard MCP transport and receives a governance verdict at each check-in. The system has been in continuous production since November 2025.

Table 2: Production deployment summary at the February 20, 2026 v5 snapshot. See Section 11.7 item 4 for provenance and reproducibility caveat.

Metric	Value
Registered agents	903
Active (> 5 updates)	75
Max updates (single agent)	48,668
Observation window	69 days

The 903 registered agents include both long-running production agents (5 agents with $>$ 1,000 updates, peak at 48,668) and shorter-lived agents from development and testing cycles. The top two agents have accumulated 48,668 and 44,529 updates respectively, providing substantial trajectory data for convergence analysis.

11.2 Observed Fleet Dynamics

For the 75 active agents with > 5 updates in the snapshot, the observed equilibrium statistics are reported in Table 3. These are pooled across the agent classes outlined in Section 2; per-class values are deferred to subsequent work, as discussed in Section 1.2 and Section 8.

Table 3: Observed fleet-aggregate EISV equilibrium (75 active agents). Per-class breakdown deferred to subsequent work.

Variable	Mean	Std	Range
E	0.766	0.087	$[0.60, 1.00]$
I	0.877	0.111	$[0.59, 1.00]$
S	0.082	0.045	$[0.01, 0.20]$
V	-0.033	0.028	$[-0.089, 0.007]$
$C(V)$	0.483	0.012	$[0.46, 0.50]$

Remark 11.1 (Reading fleet-aggregates honestly)**.** *The pooled statistics in Table 3 mix an embodied agent's sensor-driven state with session-bounded text-generation agents and resident cron-driven janitors. The means and standard deviations are real measurements and useful for cross-validation against the dynamical predictions of Section 3, but they do not characterize any single agent class accurately. A high-entropy divergent agent and a low-entropy convergent agent contribute equally to the S column; whether either is "healthy" is a class-conditional question that Section 8 formalizes and subsequent work answers empirically.*

Table 3 summarizes the observed equilibrium. These values are consistent with several qualitative predictions of the dynamics:

1. $I > E$**:** Mean integrity (0.877) exceeds mean energy (0.766), consistent with a conservative bias (Figure 1). The negative mean $V = -0.033$ (integrity surplus) is consistent with this.

2. **Low entropy:** Mean $S = 0.082$ with strong decay ($\mu = 0.8$) indicates effective uncertainty management.

3. **Coherence at 0.49:** The observed mean $C = 0.483$ (Figure 3) lies near the transform midpoint. This observed concentration does not by itself determine the information in the score or its contribution to threshold decisions (Remark 3.4).

4. V **confined to** $[-0.1, 0.1]$**:** 100% of active agents have V within $[-0.089, 0.007]$ (Figure 2). This snapshot records a narrow operating range; it does not isolate damping's causal contribution to that range.

11.3 Dynamical Regimes

Agents distribute across three dynamical regimes based on EISV trajectory trends:

The regime profiles (Figure 4) differ in integrity and entropy, though barely in coherence: convergent agents have the highest integrity and lowest entropy; divergent agents have

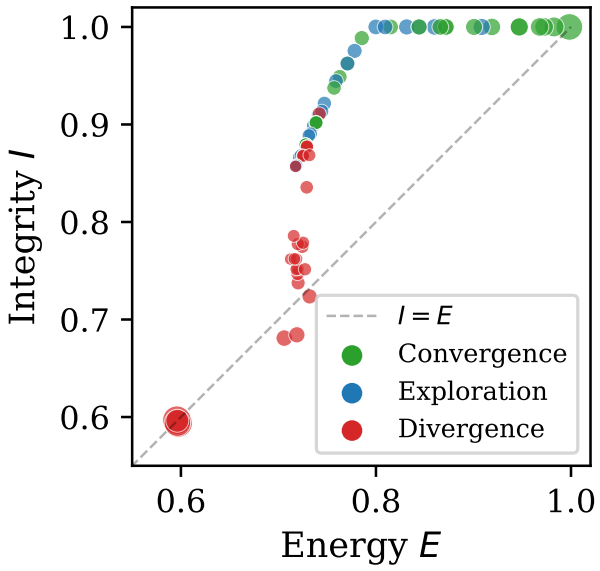


Figure 1: Energy–Integrity scatter plot for 75 active agents, colored by dynamical regime. Nearly all agents lie above the $I = E$ diagonal, consistent with a conservative bias ($\bar{V} < 0$). Point size reflects log update count.

Table 4: Regime Distribution and EISV Profiles (Active Agents)

Regime	n	$\bar{E}$	$\bar{I}$	$\bar{S}$	$\bar{C}$
Convergence	24	.838	.961	.052	.474
Exploration	22	.766	.926	.090	.477
Divergence	29	.706	.770	.100	.496

the lowest integrity and highest entropy. Coherence is slightly higher in the divergent regime ($\bar{C} = 0.496$) because these agents have smaller $|V|$, placing them nearer the coherence function's midpoint.

11.4 Operational Artifacts: Knowledge Graph and Dialectic

The governance system accumulates knowledge-graph discoveries classified by severity and type; insights, notes, and improvements dominate the corpus, with bugs and improvements forming the actionable governance output.

The dialectic protocol—a structured thesis/antithesis/synthesis process for resolving

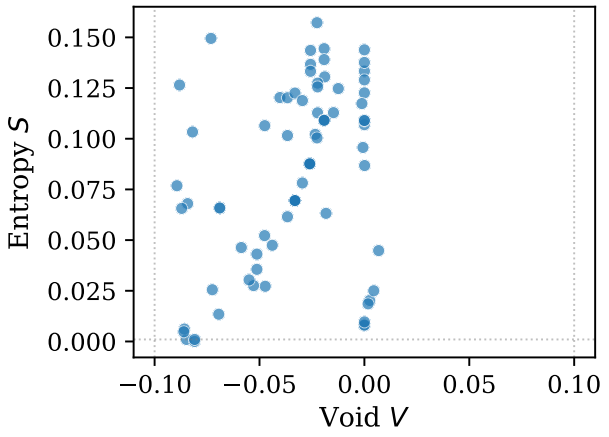


Figure 2: Entropy–Void operating region. All agents have $V \in [-0.089, 0.007]$, well within the $[-0.1, 0.1]$ predicted by damping analysis. Entropy remains low ($S < 0.20$) due to strong decay $\mu = 0.8$.

governance disputes—operates without autonomous reviewer agents in the present deployment: the resident fleet (monitors, janitors, pattern-matchers, sync daemons, sensory processes) carries no thesis-response pathway, so every antithesis to date has required human routing. Any session-level non-resolution rate observed in the audit log should therefore be read as an upper bound under facilitator load rather than as a protocol-level failure rate; a fleet with autonomous reviewers would be expected to resolve more readily.

Retrieval infrastructure (April 2026). The knowledge graph's counts above describe *storage*; whether agents could *locate* prior discoveries was a separate question. A dogfooding review in mid-April 2026 found that semantic retrieval over the discovery corpus was returning similarity scores near the noise floor across diverse queries, tag filters were post-filter rather than ranking signals, and an unlabeled low-threshold fallback was returning random-looking results as if they were matches. The retrieval layer was rebuilt in a five-phase sequence (2026-04-20): an honest-failure-mode fix, a reproducible eval harness, a modern embedder (`BAAI/bge-m3`, 1024-dimensional), and hy-

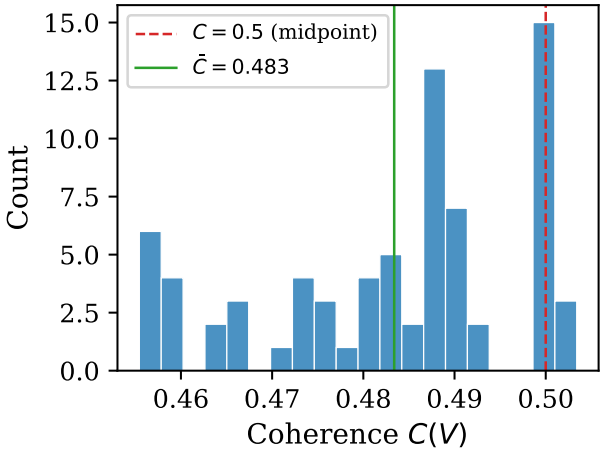


Figure 3: Distribution of coherence $C(V)$ across active agents. The mean $\bar{C} = 0.483$ lies near the theoretical midpoint ($C = 0.5$), consistent with small $|V|$. The narrow spread means coherence barely separates agents.

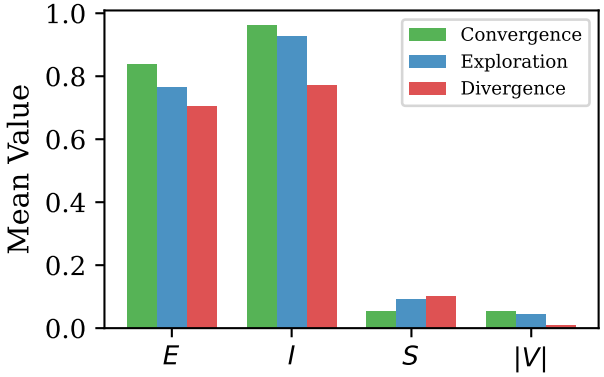


Figure 4: Mean EISV component values by dynamical regime. Convergent agents show the highest I and lowest S; divergent agents show the reverse pattern. The $|V|$ is small across all regimes, confirming effective damping.

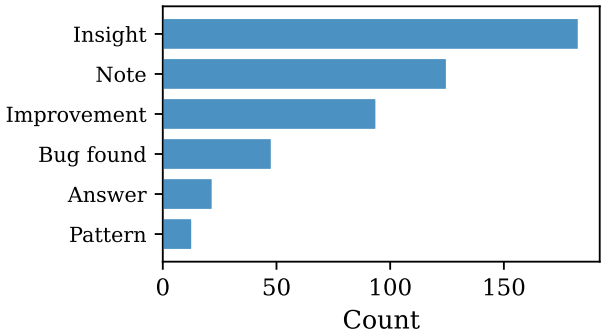


Figure 5: Knowledge graph discovery classification by type. Insights and notes dominate, with bugs and improvements constituting the actionable governance outputs.

brid retrieval via reciprocal rank fusion over BM25 and dense scores. Cross-encoder reranking and 1-hop typed-edge expansion were landed as opt-in modes; on the current corpus size they showed no aggregate gain. The rebuild is prerequisite to any empirical claim that relies on agents *using* prior discoveries rather than merely *storing* them; documentation and baselines live in the governance repository under `docs/plans/2026-04-20-kg-retrieval-rebuild.md`.

11.5 Phase 2 Measurement: Per-Class Envelopes

The Phase 2 calibration protocol of Section 8 was executed against a 30-day healthy-regime slice of `core.agent_state` on April 18, 2026. Table 5 reports the resulting per-class healthy operating points and manifold radii. The script that produced these values (`scripts/calibrate_class_conditional.py`) ships with the implementation; the values themselves are now committed in `config/governance_config.py` as the production class-conditional constants, replacing the Phase 1 placeholders for the manifold-coherence path.

Two observations from Table 5 are worth surfacing.

First, the per-class manifold radii span a 3.3× range (Lumen 0.119 to Watcher 0.395). The Phase 1 placeholder of $\|\Delta\|_{\max} = 1.8$ was therefore 4 to 15× larger than the empirically measured envelope of any class. A fleet-wide normalization at the placeholder value would have produced a manifold coherence saturating near 1.0 for every agent regardless of state—essentially a constant signal. This is the quantitative motivation behind the homogenization argument of Section 2: an overly broad normalization radius can reduce within-class score variation. Whether a measured fleet-wide radius performs worse than

Table 5: Phase 2 measured per-class scale constants from a 30-day healthy-regime slice (April 18, 2026). $(E, I, S)_{\text{healthy}}$ is the median; $\|\Delta\|_{\max}$ is the 95th percentile of state-space distance from each class's own healthy operating point. Five classes with $N \geq 30$ qualified for measurement; classes below the threshold fall back to the fleet-wide default.

Class	**N**	E_h	I_h	S_h	$\|\Delta\|_{\max}$
Lumen	7320	0.745	0.800	0.168	0.119
default	2033	0.726	0.793	0.236	0.202
Sentinel	1870	0.751	0.798	0.193	0.170
Vigil	384	0.737	0.790	0.240	0.171
Watcher	283	0.748	0.769	0.248	0.395

the class-specific radii requires a separate comparison.

Second, the healthy operating points cluster more tightly on E and I than on S. Across the five measured classes, $E_h \in [0.726, 0.751]$ and $I_h \in [0.769, 0.800]$, but $S_h \in [0.168, 0.248]$—a wider spread on entropy. This matches the qualitative expectation that what makes classes behaviorally distinct is largely how much uncertainty they operate under: Lumen at $S_h = 0.17$ runs the most convergent; Watcher at $S_h = 0.25$ runs more divergent because its work (pattern-match over recently-edited code) is intrinsically exploratory.

Table 5 replaces the Phase 1 placeholder values for the manifold-coherence path; the remaining $S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}$ constants (used only by the not-yet-shipped tier-1 logprob path and tier-2 multi-sample path) remain as Phase 1 placeholders.

11.6 Verdict Counterfactual

The grounded manifold-coherence form (Equation (13)) takes different inputs than the legacy operational form (Equation (5)): the grounded form computes a class-conditional distance from $(E, I, S)_{\text{healthy}}(c)$, while the legacy form maps V through tanh. The choice of coherence formula can therefore change basin labels, not just the raw value reported. We quantify this directly by re-running the basin-classification step on a counterfactual in which the grounded form replaces the legacy form on the same production state vectors.

Table 6: Verdict counterfactual: per-class basin flip rates when the grounded class-conditional manifold coherence replaces the legacy fleet-wide tanh form on the same state vectors.

Class	**N**	**Flips**	**%**	$\|\Delta C\|_{\max}$
Lumen	7889	2548	32.3	0.51
default	2358	722	30.6	0.50
Sentinel	2227	351	15.8	0.51
Vigil	472	156	33.1	0.50
Watcher	364	67	18.4	0.48
Total	**13310**	**3844**	**28.9**	

The basin classifier `classify_basin` maps (E, I, S, V, C, R) to one of three regions—`high`, `boundary`, `low`—consumed by the production dashboard and escalation logic alongside the Φ-thresholded verdict of Equation (21). A *basin flip* is a row whose classifier output differs between the legacy and grounded coherence inputs; it is the observable downstream of which this counterfactual measures.

Setup. For each row in a 30-day window of `core.agent_state` ($N = 13{,}310$ rows with computable E), we compute C_{legacy} (the value already stored in the row, produced by the production server using Equation (5)) and $C_{\text{grounded}}(c)$ (the manifold form Equation (13) with class-conditional $(E, I, S)_{\text{healthy}}(c)$ and $\|\Delta\|_{\max}(c)$ from Table 5, where c is determined by the identity-tag classifier of Section 2.2). Each row is classified into a basin twice using the same `classify_basin(`E, I, S, V, C`,risk)` function used in production. A flip is a row whose basin assignment differs.

Result. 3,844 of 13,310 rows (28.9%) flip basin assignment. Table 6 reports per-class breakdown and Table 7 reports flip directions.

Interpretation. The formula change is not a cosmetic relabeling even when both quantities live in $[0, 1]$: it changes basin labels for 15–33% of rows per class. The legacy form is nearly constant (about 0.49 in the reported deployment; Section 11). The replacement changes

Table 7: Flip-direction breakdown by class. Predominance of `high→low` and `boundary→low` indicates the two formulations disagree asymmetrically: state vectors the fleet-wide form classifies as healthy are more often reclassified as breaching under the class-conditional form than vice-versa.

Class	high→low	bnd→low	high→bnd
Lumen	1169	758	621
default	140	518	64
Sentinel	35	289	27
Vigil	70	70	16
Watcher	0	67	0

labels where its values cross the classifier's operative thresholds; variation alone does not imply a basin change. This comparison combines a change from a fleet-wide tanh of V to a distance-based score with class-conditional calibration; it does not isolate their separate contributions.

The dominant transition is into the `low` basin: state vectors labeled healthy or borderline by the historical classifier are reclassified as breaching a bound under the replacement score. The magnitude (29% fleet-wide, up to 33% per class) describes this paired label comparison. Its directional asymmetry depends on the two formulas, the observed states, and the classifier thresholds; it is not independent evidence that class-conditional calibration improves governance. Isolating that contribution would require a fleet-wide distance-score baseline with the same inputs and formula.

The script replays the basin classifier, not the complete decision pipeline or its controller history. A flip into `low` therefore does not establish that an agent would receive a `guide` or `pause` intervention. Neither label set has outcome grades in this comparison. Whether the replacement helps operators choose interventions remains an empirical question.

Data availability and reproducibility. The public reproducibility kit archives two CSV exports and an offline counting script. Its pinned repository revision contains `analysis.py`; it reads the archived rows without a database. The submission export contains 13,292 rows and 3,831 stored-label flips (28.8%), compared with the original table's 13,310 rows and 3,844 flips. These are distinct pulls; the export does not exactly reproduce the table, and the missing rows cannot be reconstructed from the export. We retain the historical table rather than silently substitute the nearby pull. Recomputing scores from rounded exported coordinates is a further check, distinct from counting stored labels.

The second archived window, ending April 23, contains 16,879 rows and 7,476 stored-label flips (44.3%) under the frozen calibration constants. This is a different, overlapping observation window, not an independent replication or evidence of a stable intervention benefit. It demonstrates the window dependence of the descriptive percentage.

The production database no longer retains the full April windows, as documented in the kit's August 14 retention check. The live-query script is provenance only: rerunning it may return fewer rows without an error and cannot recover the historical result. No exact raw archive of the calibration's healthy-regime input is supplied here either; the printed constants are the record of that measurement.

11.7 Limits of the Snapshot

The production data has six limitations that bound the strength of our empirical claims:

1. **Parameter evolution:** Dynamics parameters, governor thresholds, and the I-channel mode were refined iteratively during deployment. The reported EISV statistics are a snapshot across agents that experienced different parameter regimes. A controlled A/B comparison between logistic and linear modes has not been conducted.

2. **Governor recency:** The adaptive governor (CIRS v2) was deployed recently; the audit log does not yet contain sufficient pre/post data for a rigorous before/after comparison.

3. **Population heterogeneity:** The 903 agents include long-running production

agents, short-lived test agents, and agents from early development with since-changed parameters. We report results for the 75 agents with > 5 updates to mitigate this, but acknowledge that even this filtered set spans multiple configuration epochs.

4. **Temporal seam:** Fleet-aggregate statistics (Table 3) are from a February 20, 2026 snapshot; per-class envelopes (Section 11.5) and the verdict counterfactual (Section 11.6) are from a 30-day window ending April 18, 2026, during which the population grew from 903 to 3,286 registered identities. The February 20 snapshot was taken against the v5 SQLite governance store, which was later retired in the PostgreSQL migration; the snapshot is preserved as historical record and is not reproducible against the current schema. Per-class envelopes and the verdict counterfactual were run against the post-migration PostgreSQL store. Those live rows have since expired; the public CSVs preserve nearby counterfactual exports, not the exact calibration input or original table (see the data-availability statement in Section 11.6).

5. **Identity-system maturity:** The measurement window includes material revisions to the identity-resolution layer that landed in mid-to-late April 2026—removal of name-claim lookup, adoption of UUID-direct dispatch, and receipt-based onboarding authentication. Class assignment is tag-based on the identity record (Section 2.2), so an identity whose tags were inherited from a cached binding or from an archived predecessor may appear in a class whose envelope does not match its actual operational regime. We expect this to affect a bounded fraction of rows rather than the aggregate direction of the per-class envelopes, but we have not filtered the counterfactual to post-fix identities; a re-measurement on identity-clean data, once sufficient post-fix history has accumulated, is deferred to subsequent work.

6. **Retrieval infrastructure and write-path hygiene:** Two infrastructure defects predating the measurement window affect how knowledge-graph activity and dialectic resolution statistics in the audit log should be read. First, until the April-2026 retrieval rebuild (Section 11.4), semantic retrieval scored at the noise floor and cross-agent learning was effectively gated by agents knowing a-priori where to look; fleet-learning claims that depend on agents *using* prior discoveries are therefore operationally bounded by that pre-rebuild era. Second, the note-write handler silently auto-tagged every non-permanent discovery with `ephemeral` until PR #54 (2026-04-19), which scheduled those notes for 7-day auto-archive; the tag-lifecycle counts in the archived tier thus contain both agent-chosen and handler-injected values in the pre-fix window. Neither defect affects the grounded EISV counterfactual of Section 11.6 or the per-class envelopes of Section 11.5, whose inputs are `core.agent_state` rows rather than KG content; we flag them here because they bound operational claims about the shared-memory surface.

12 Re-Grounding a Deployed Framework

The counterfactual of Section 11.6 was computable because the framework was instrumented during a live migration: grounded and legacy coherence were computed on every production check-in over the measurement window, and governance gating continued to read the legacy values so no verdict changed during measurement. This section documents the mechanism that made the comparison possible. We believe it generalizes to any deployed agent framework whose coordinate semantics need to shift while service continues.

12.1 Three-Phase Dual-Compute

The migration runs in three discrete phases:

Phase 1: Dual-compute. Every check-in computes both the grounded value and the legacy value. Grounded values land in canonical $\{E, I, S, C\}$ slots; legacy values are preserved in $\{E_{\text{legacy}}, I_{\text{legacy}}, S_{\text{legacy}}, C_{\text{legacy}}\}$. Source annotations on each grounded coordinate (`e_source`, `i_source`, `s_source`, `coherence_source`) identify which computation tier produced each value (logprob / multisample / heuristic / resource / manifold / KL / FEP).

Phase 2: Calibration. Scale constants and class-conditional healthy operating points are measured against a reference corpus of healthy production agent-turns, partitioned by the identity tags of Section 2.2. Divergence between grounded and legacy values is characterized; the counterfactual of Section 11.6 is the first such characterization.

Phase 3: Migration. Governance gating (verdicts, basin classification, risk thresholds) switches from legacy to grounded values. The legacy fields remain readable for one minor version as a comparison surface, then are dropped.

The contribution of this section is not the three-phase shape itself—which is standard for any deprecated-API migration—but the mechanism by which Phase 1 preserves governance behavior while exposing grounded values to downstream consumers.

12.2 Pipeline-Ordering as a Behavior-Preservation Mechanism

A naive reading of Phase 1 is that grounded values must equal legacy values for behavior to be preserved. This is false in general: the grounded coherence C under Equation (13) differs numerically from the legacy tanh-based coherence even when computed on identical EISV inputs, because the formulas are different. Requiring numerical equality would defeat the point of grounding.

The mechanism we use instead is pipeline-ordering. The server's `process_agent_update` request pipeline runs in two stages—which we will call the *decision stage* (internally named `phases` in the server code) and the *payload stage* (internally `enrichments`). The decision stage computes governance state and issues the verdict; the payload stage augments the response with secondary signals. Payload-stage handlers are registered with integer priorities in the range $[0, 100]$ (lower runs first). We register the grounding computation as a payload-stage handler at priority 75, downstream of every handler that contributes to the verdict and basin classification but upstream of presentation handlers such as the human-readable summary and the broadcaster event.

The result: by the time the grounding handler runs, the verdict has already been computed—using the legacy values produced by the decision stage. The grounding handler then swaps grounded values into the canonical $\{E, I, S, C\}$ slots, copying the originals into $\{E_{\text{legacy}}, \dots\}$. The response payload carries grounded values, but the governance decision was made on legacy. No gating code is touched; no consumer reading $\{E, I, S, C\}$ sees a missing field; no decision changes during Phase 1.

Remark 12.1 (Scope of Phase 1)**.** *V is not dual-computed; its grounded reinterpretation as accumulated free-energy debt (Equation (11)) is a relabeling, not a new computation. Tier-1 (logprob) and tier-2 (multisample) grounding paths are interface stubs that raise `NotImplementedError`, with the dispatcher falling through to tier-3; inference-layer instrumentation for tier-1 lands separately.*

12.3 Generalization

The pipeline-ordering mechanism is reusable for any deployed agent framework facing a semantic-coordinate change while service continues. The required structure is: (i) a request pipeline with explicit ordering between decision-producing stages and presentation-producing stages; (ii) a canonical payload slot for the co-

ordinate being re-grounded; (iii) identity (or session) keying that allows audit consumers to tag rows by the migration phase active when the row was produced. Frameworks that conflate decision and presentation into a single non-ordered handler must first separate the two concerns—a separation worth doing independently of any migration, because it makes governance decisions auditable and presentation choices revisable in isolation.

As a concrete second example, consider an RLHF-governed deployment whose gating signal is a reward-model score $r(x, y)$ and whose operators wish to migrate to a calibrated reward $\tilde{r}(x, y)$ derived from a revised preference model without changing refusal or escalation behavior during the transition. The pipeline structure maps directly: the *phases* stage computes the legacy r and issues the gate; the *enrichments* stage computes $\tilde{r}$, writes it to a canonical `reward` field, and copies r to `reward_legacy`. Downstream dashboards and calibration pipelines see $\tilde{r}$; refusal logic continues to read r. The mechanism is identical; only the coordinate being regrounded changes.

13 Discussion

13.1 Design Principle: Adjust Expectations, Not Measurements

A core design principle of UNITARES is that observations are never transformed to appear healthier. When production coherence averages $C \approx 0.49$ rather than the "expected" $C \approx 0.7$, the system reports 0.49. The phase-aware governor (Section 6) addresses the gap by adjusting *thresholds* (expectations), not *signals* (measurements).

This principle—**adjust expectations, not measurements**—has three practical consequences:

1. **Calibration integrity:** Drift detection (Section 7) compares raw signals against baselines. Inflating measurements would mask the very drift the system is designed to detect.
2. **Operator trust:** Dashboard values correspond to actual system state. An operator seeing $C = 0.49$ knows it is the raw value, not a rescaled one; its decision contribution still depends on the operative thresholds.
3. **Model provenance:** The certificate in Section 5 concerns its specified vector field. A presentation-only transformation does not change that field; feeding transformed measurements back into it requires analyzing the resulting dynamics.

13.2 Parameter Sensitivity

The system's behavior is most sensitive to four parameters:

Convergence rate α. Controls how quickly E tracks I. Production uses $\alpha = 0.42$. Increasing α strengthens both the E-channel diagonal damping and its coupling from I; any claim about the contraction rate must check the complete matrix inequality.

Coherence coupling β_I. Determines how strongly coherence restores integrity. The reported system uses $\beta_I = 0.30$. Its derivative contribution is $\beta_I C'_{\max} = 0.15$; comparison with one diagonal entry is insufficient. The complete row conditions in Appendix B certify the printed parameter set.

Self-regulation γ_I. Determines the I-channel's intrinsic damping. In logistic mode, $\gamma_I = 0.25$ yields negative m_{sat} at the reported operating points (Appendix C), indicating forcing above maximum logistic damping in that snapshot. In linear mode, $\gamma_I = 0.169$ was tuned to give $I^* \approx 0.80$, well within $(0, 1)$ for that forcing level. Other forcing levels can still produce boundary contact.

Void damping δ. At $\delta = 0.4$, the uncoupled relaxation timescale is $\tau_V = 1/\delta = 2.5$ in model time units. For a fixed input gap $d = E - I$, the scalar channel has $V^* = \kappa d/\delta$. The observed range depends on that input trajectory as well

as damping (Remark 3.4); δ alone does not confine V to the observed interval. A smaller δ increases this fixed-input gain and lengthens the relaxation time.

13.3 Verdict Sensitivity to the Objective Weights

The scalar governance signal of Equation (7) uses default weights $w_E = w_I = w_S = w_V = w_\eta = 0.5$. The equal-weight choice is a deployment default, not a derived optimum; an honest treatment asks how sensitive the verdict surface is to weight perturbations.

At the verdict boundary $\Phi = 0.15$, the partial derivatives against each weight are the corresponding state-dependent contributions:

$$\frac{\partial \Phi}{\partial w_E} = E, \qquad \frac{\partial \Phi}{\partial w_I} = -(1 - I), \quad \frac{\partial \Phi}{\partial w_S} = -S,$$
$$\frac{\partial \Phi}{\partial w_V} = -\left|V\right|, \quad \frac{\partial \Phi}{\partial w_\eta} = -\left\|\Delta\boldsymbol{\eta}\right\|^2.$$

Evaluating at the production fleet-aggregate operating point $(E, I, S, V) = (0.77, 0.88, 0.08, -0.03)$ with healthy $\|\Delta\boldsymbol{\eta}\|^2 \lesssim 0.01$:

Weight	$\partial\Phi/\partial w_i$	Sign
w_E	$+0.77$	promotes proceed
w_I	-0.12	promotes proceed (small; I high)
w_S	-0.08	promotes proceed (small; S low)
w_V	-0.03	negligible
w_η	$\lesssim 0.01$	negligible

The w_E derivative dominates by an order of magnitude at healthy operating points. This is structural rather than incidental: E sits near 0.77 in production while the penalty terms are all near zero. The objective is therefore most sensitive to the energy weight and least sensitive to the void/drift weights *in the healthy regime*. The ordering flips at degraded states where S rises and $(1 - I)$ grows.

Threshold robustness band. Perturbing all five weights uniformly by $\pm\epsilon$ shifts Φ by at most $\epsilon \cdot (E + (1-I) + S + |V| + \|\Delta\boldsymbol{\eta}\|^2) \approx 1.01\epsilon$ in magnitude at the production operating point. The verdict boundary $\Phi = 0.15$ is therefore robust to uniform weight perturbations up to roughly $\pm 15\%$ before a single-agent verdict at the boundary would flip. Non-uniform perturbations are bounded by the largest single partial derivative; a $+50\%$ perturbation to w_E alone would shift Φ by $+0.19$ at the operating point, which is well above 0.15 and would change boundary-case verdicts.

What this analysis does not do. The derivatives above characterize local sensitivity at a single operating point; they do not establish how many verdicts would flip under a specified weight perturbation, nor whether equal weights are in any sense optimal. That result requires an empirical counterfactual: rerun the audit log with perturbed weights and count verdict flips, analogous to the grounded-form counterfactual reported in Section 11. We defer the weight-perturbation verdict-flip study to subsequent work with the same infrastructure.

13.4 Limitations

1. The deployed coordinates are computed by tier-3 heuristic estimators; the information-theoretic definitions of Section 4 are target semantics, and the tier-1 and tier-2 estimators are not implemented. No result in this paper grades the coordinates or either coherence form against independently labelled outcomes.

2. The coherence function $C(V, \Theta)$ creates a nonlinear feedback loop whose gain depends on the operating point (maximum at $V = 0$, vanishing at $|V| \gg 0$). The contraction analysis uses $C'_{\max}$ as a worst-case bound for the fixed continuous model. It does not include state-dependent input derivatives, controller history or the implemented finite-step map.

3. The behavioral–epistemic drift vector's four components are currently equally weighted; adaptive weighting based on domain context is future work.

4. The multi-agent network analysis assumes fixed topology; production agent populations are dynamic with agents joining and leaving.

5. The PID governor gains (K_p, K_i, K_d) were tuned empirically; optimal gain selection via, e.g., Ziegler-Nichols or LQR methods is future work.

6. The empirical program of Section 11 was computed over a window that overlaps with material identity-system revisions (Section 11.7, item 5). A replication of the per-class calibration (Section 11.5) and the verdict counterfactual (Section 11.6) on identities that post-date those revisions is the natural validation path and is deferred to subsequent work.

14 Conclusion

This paper presented two structural reframings of UNITARES. First, the EISV state vector is given information-theoretic target semantics: each coordinate is specified as a Shannon entropy, mutual information, variational free energy, or KL/manifold distance, to be measured by a tiered computation recipe and normalized by a provenance-tagged scale constant. The deployed system computes all four coordinates with the tier-3 heuristic estimators; the higher tiers are specified but not implemented. Second, fleet-wide normalization is rejected as the wrong target; scale constants and healthy operating points become functions of an agent class, keyed on existing identity tags. Together these define a path from a phenomenological dynamical system to a class-conditional framework whose coordinates could be derived rather than guessed. This paper specifies that path; it does not complete it.

The pipeline-ordering migration mechanism (Section 12.2) is a methodological contribution beyond UNITARES itself: any deployed agent framework that needs to shift its semantic coordinates while preserving live governance behavior can apply the same separation between decision-producing and presentation-producing pipeline stages. We argue this separation is worth doing on its own merits, independent of any specific migration.

The corrected certificate in Appendix B proves continuous incremental contraction for the printed fixed parameter set while trajectories remain in its domain under common exogenous inputs. It does not establish domain invariance, an interior equilibrium or contraction of the deployed finite-step controller. Grounded coherence changes the field and requires a new certificate (Remark 5.1). Production deployment statistics from 903 agents over a 69-day observation window appear in Section 11; Section 11.5 reports a first class-conditional measurement for the manifold-coherence path, and Section 11.6 shows that 28.9% of basin assignments flip under the historical counterfactual. This is a label comparison, not a measurement of intervention quality; the nearby public export and its limits are documented in that section. The remaining scale constants and higher-tier estimators are deferred to subsequent work following the measurement protocol of Section 8.

The framework's adaptive governor (Section 6), behavioral–epistemic drift vector (Section 7), and multi-agent extensions (Section 10) form the rest of the governance surface alongside these two commitments.

The current verdict surface—`proceed`, `guide`, `pause`, `reject`—is threshold-triggered from the CIRS v2 control law (Section 6), not derived from the minimization of an expected free energy. A principled extension under active inference Parr et al. [2022], Da Costa et al. [2020] would recast governance interventions as policy selections that minimize expected free energy over future trajectories; the current framework monitors free-energy proxies and intervenes via phase-conditioned thresholds. We do not claim the existing verdicts are active-inference actions, and the reframing is future work.

A cleaner identification of the deferred step: Tier-1 computation of $-F$ (Section 4.2) requires an explicit generative model $p(z, o)$ of the agent. For closed-weights frontier agents, it is not obvious what that generative model *is* in the strict variational sense. The weights are not directly

accessible; a tokenized posterior over next tokens is a tractable but coarse surrogate; a semantic generative model over tool-use trajectories is richer but speculative. Which choice, if any, supports a faithful $-F$ estimator is an open methodological problem, not an engineering deferral. The natural empirical test is whether a candidate $\hat{F}$ adds predictive value over the current EISV proxies; that horse race is left to subsequent work.

As autonomous AI agents become more prevalent, governance frameworks must accommodate the heterogeneity of the populations they govern—embodied creatures, persistent observers, session-bounded assistants, ephemeral parsers—without collapsing them to a homogeneous distribution. UNITARES v6 presents one structural commitment to that requirement.

References

W. Lohmiller and J.-J. E. Slotine. On contraction analysis for non-linear systems. *Automatica*, 34(6):683–696, 1998.

K. Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

L. Da Costa, T. Parr, N. Sajid, S. Veselic, V. Neacsu, and K. Friston. Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology*, 99:102447, 2020.

N. Sajid, P. J. Ball, T. Parr, and K. J. Friston. Active inference: demystified and compared. *Neural Computation*, 33(3):674–712, 2021.

R. Smith, K. J. Friston, and C. J. Whyte. A step-by-step tutorial on active inference and its application to empirical data. *Journal of Mathematical Psychology*, 107:102632, 2022.

T. Parr, G. Pezzulo, and K. J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.

C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

J.-J. E. Slotine. Modular stability tools for distributed computation and control. *International Journal of Adaptive Control and Signal Processing*, 17(6):397–416, 2003.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *ICML*, pages 1321–1330, 2017.

A. Niculescu-Mizil and R. Caruana. Predicting good probabilities with supervised learning. In *ICML*, pages 625–632, 2005.

K. J. Åström and R. M. Murray. *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press, 2008.

Y. Bai, S. Kadavath, S. Kundu, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

L. Ouyang, J. Wu, X. Jiang, et al. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022.

T. Rebedea, R. Dinu, M. Sreedhar, C. Parisien, and J. Cohen. NeMo Guardrails: A toolkit for controllable and safe LLM applications with programmable rails. In *EMNLP (Demo)*, 2023.

S. Ravindran. Moral Anchor System: A predictive framework for AI value alignment and drift prevention. *arXiv preprint arXiv:2510.04073*, 2025.

L. Kuhn, Y. Gal, and S. Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*, 2023. arXiv:2302.09664.

C. L. Wang, T. Singhal, A. Kelkar, and J. Tuo. MI9: An integrated runtime governance framework for agentic AI. *arXiv preprint* arXiv:2508.03858v4, 2025.

A. Rath. Agent drift: Quantifying behavioral degradation in multi-agent LLM systems over extended interactions. *arXiv preprint arXiv:2601.04170*, 2026.

A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada. Control barrier functions: Theory and applications. In *European Control Conference (ECC)*, pages 3420–3431, 2019.

Q.-C. Pham, N. Tabareau, and J.-J. E. Slotine. A contraction theory approach to stochastic incremental stability. *IEEE Transactions on Automatic Control*, 54(4):816–820, 2009.

K. Itô. Stochastic integral. *Proceedings of the Imperial Academy*, 20(8):519–524, 1944.

H. Wang, C. M. Poskitt, J. Wei, and J. Sun. ProbGuard: Proactive runtime monitoring for LLM agent safety via probabilistic prediction. In *ASE*, pages 1470–1482, 2026. doi:10.1145/3832783.3837413.

A Parameter Tables

Table 8: EISV parameter values used in this paper; distinct from current active defaults (Section 5).

Parameter	Symbol	Val.	Role
Energy channel			
Convergence	α	0.42	E tracks I
Entropy erosion	β_E	0.10	S erodes E
Drift excitation	γ_E	0.05	Drift raises E
Information channel			
Entropy degrad.	k	0.10	S degrades I
Coherence boost	β_I	0.30	$C(V)$ restores I
Self-regulation	γ_I	0.25	Logistic damping
Linear mode	γ_I^{lin}	0.169	Linear damping
Entropy channel			
Natural decay	μ	0.80	S relaxation
Drift injection	λ_1	0.30	Base (adaptive)
Coherence red.	λ_2	0.05	$C(V)$ reduces S
Complexity	β_c	0.15	Task $\rightarrow$ entropy
Noise	σ	0.02	Assumed noise
Void channel			
Coupling	κ	0.30	E-I gap
Damping	δ	0.40	V relaxation
Coherence			
Maximum	$C_{\max}$	1.0	Upper bound
Sensitivity	C_1	1.0	Tanh slope

Table 9: CIRS v2 Adaptive Governor Parameters

Parameter	Symbol	Val.	Role
Proportional	K_p	0.05	Gentle corr.
Integral	K_i	0.005	Drift corr.
Derivative	K_d	0.10	Damping
Integral limit	$I_{\max}$	0.10	Wind-up cap
Reference points			
Expl. τ ref	$r_\tau^{(e)}$	0.35	Forgiving
Expl. β ref	$r_\beta^{(e)}$	0.55	Forgiving
Intg. τ ref	$r_\tau^{(i)}$	0.40	Strict
Intg. β ref	$r_\beta^{(i)}$	0.60	Strict
Safety bounds			
τ floor/ceil	τ	0.25/0.75	Hard bounds
β floor/ceil	β	0.20/0.70	Hard bounds

Table 10: Behavioral–Epistemic Drift Vector Parameters

Parameter	Symbol	Val.	Role
Baseline tracking			
EMA smoothing	α_{ema}	0.1	Update rate
Warmup steps	n_w	5	Drift suppression
Stability measurement			
Decision win.	w_s	20	Consistency
Transition pen.	—	/flip	Instability
Coupling to dynamics			
Energy excit.	γ_E	0.05	Drift $\rightarrow dE$
Entropy inj.	λ_1	.05–.20	Drift $\rightarrow dS$

Table 11: Grounding scale constants (Section 4.3). The manifold-coherence path is now measured class-conditionally in Table 5; the remaining constants below remain Phase 1 placeholders pending tier-1/tier-2 calibration work.

Constant	Sym.	Val.	Pctl	Prov.
Entropy	S_{scale}	3.0	—	placeholder
MI	I_{scale}	2.0	—	placeholder
Free energy	E_{scale}	1.0	—	placeholder
Manifold rad.	$\|\Delta\|_{\max}(c)$	per c	95	measured
Tok./sec env.	$(\text{t/s})_{\max}$	200	—	placeholder

The Pctl column records the percentile basis once the constant has been measured against a healthy corpus (per Section 8); the Provenance column is one of {`placeholder`, `measured`, `derived`} per Definition 8.2. The $\|\Delta\|_{\max}(c)$ row is now measured class-conditionally in Table 5; the remaining entries are still `placeholder` scaffolding awaiting the tier-1/tier-2 calibration work. A later revision can restore the date and corpus-size columns once those remaining constants are measured.

B Conditional Continuous Contraction Certificate

This appendix proves an incremental bound for the printed linear-I continuous vector field. It is distinct from a proof for the numerical integrator or the history-dependent governor.

Domain, Inputs and Jacobian

Let $\mathcal{D} = [0,1]^2 \times [S_{\min}, 1] \times [-1, 1]$, $S_{\min} = 0.001$. Parameters are fixed and nonnegative, with $\alpha, \gamma_I, \mu, \delta > 0$. The common Θ is fixed, as are $C_{\max}$, C_1, λ_1 and λ_2. Drift and complexity are exogenous inputs shared between compared trajectories and may vary with time. Noise is absent in this continuous deterministic certificate. State-dependent input feedback would contribute additional derivatives.

For a constant positive diagonal metric $M = \operatorname{diag}(m_E, m_I, m_S, m_V)$, the sufficient uniform matrix condition is

$$F(x) = MJ(x) + J(x)^\top M \preceq -2aM, \qquad x \in \mathcal{D}, \qquad a > 0. \tag{34}$$

The Jacobian of (1)–(4), with linear g_I, is

$$J = \begin{pmatrix} -\alpha - \beta_E S & \alpha & -\beta_E E & 0 \\ 0 & -\gamma_I & -k & \beta_I q \\ 0 & 0 & -\mu & -\lambda_2 q \\ \kappa & -\kappa & 0 & -\delta \end{pmatrix}, \tag{35}$$

where $q = C'(V) = C_{\max} C_1 \operatorname{sech}^2(C_1 V)/2$ and $0 \le q \le q_{\max} = C_{\max} C_1/2$. This interval may conservatively enlarge the derivative range on $\mathcal{D}$.

Explicit Sufficient Conditions

The diagonal entries of F are

$$F_{11} = -2m_E(\alpha + \beta_E S), \tag{36}$$
$$F_{22} = -2m_I \gamma_I, \tag{37}$$
$$F_{33} = -2m_S \mu, \tag{38}$$
$$F_{44} = -2m_V \delta. \tag{39}$$

Its nonzero off-diagonal entries, with $F_{ji} = F_{ij}$, are

$$F_{12} = m_E \alpha, \tag{40}$$
$$F_{13} = -m_E \beta_E E, \tag{41}$$
$$F_{14} = m_V \kappa, \tag{42}$$
$$F_{23} = -m_I k, \tag{43}$$
$$F_{24} = m_I \beta_I q - m_V \kappa, \tag{44}$$
$$F_{34} = -m_S \lambda_2 q. \tag{45}$$

An off-diagonal entry's sign alone does not determine its effect on a quadratic form. The following bounds use absolute values, retaining the cancellation within F_{24}.

Define

$$B_{24} = \max_{q \in \{0, q_{\max}\}} |m_I \beta_I q - m_V \kappa|,$$
$$B_4 = \max_{q \in \{0, q_{\max}\}} \big(|m_I \beta_I q - m_V \kappa| + m_S \lambda_2 q\big).$$

Both maxima occur at interval endpoints because their arguments are convex functions of q. Uniform upper bounds for the row sums of $F + 2aM$ are

$$\begin{aligned} R_E = & -2m_E(\alpha + \beta_E S_{\min}) + 2am_E \\ & + m_E \alpha + m_E \beta_E + m_V \kappa, \end{aligned} \tag{46}$$
$$\begin{aligned} R_I = & -2m_I \gamma_I + 2am_I + m_E \alpha \\ & + m_I k + B_{24}, \end{aligned} \tag{47}$$
$$\begin{aligned} R_S = & -2m_S \mu + 2am_S + m_E \beta_E \\ & + m_I k + m_S \lambda_2 q_{\max}, \end{aligned} \tag{48}$$
$$R_V = -2m_V \delta + 2am_V + m_V \kappa + B_4. \tag{49}$$

Here $E \le 1$ and $S \ge S_{\min}$ are used explicitly.

Theorem B.1 (Conditional EISV Contraction)**.** *Under the domain, parameter and input assumptions above, if some positive diagonal M and $a > 0$ satisfy*

$$R_E \le 0, \quad R_I \le 0, \quad R_S \le 0, \quad R_V \le 0, \tag{50}$$

then any two solutions that remain in $\mathcal{D}$ on $[0,T]$ satisfy

$$\|x_1(t)-x_2(t)\|_M \le e^{-at}\|x_1(0)-x_2(0)\|_M, \quad 0\le t\le T. \tag{51}$$

Proof. The row bounds imply by the symmetric Gershgorin theorem that $F+2aM \preceq 0$ throughout $\mathcal{D}$. For $d = x_1 - x_2$, convexity keeps the segment $x_2 + ud$, $0 \le u \le 1$, inside $\mathcal{D}$. With the inputs held common,

$$f(x_1,t) - f(x_2,t) = \int_0^1 J(x_2+ud,t)d\,du.$$

Consequently $\frac{d}{dt}(d^\top M d) \le -2ad^\top M d$. Integration and a square root give the bound. □

These four row conditions do not establish domain invariance.

Fixed-Parameter Verification

Remark B.2 (Printed Certificate Parameters)**.** *The certificate uses $\alpha = 0.42$, $\beta_E = 0.10$, $\beta_I = 0.30$, $\gamma_I = 0.25$, $k = 0.10$, $\mu = 0.80$, $\kappa = 0.30$, $\delta = 0.40$, $\lambda_2 = 0.05$, $q_{\max} = 0.5$ and $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$. These are the printed proof parameters, not a claim that they match a current deployment's active parameter set.*

For $a = 0.019$, $B_{24} = 0.024$ and $B_4 = 0.031$. Exact rational substitution in the four inequalities gives:

Row	Exact upper bound	Decimal
E	$-211/50000$	-0.00422
I	$-4/625$	-0.0064
S	$-1507/1000$	-1.507
V	$-149/25000$	-0.00596

All four are strictly negative. Thus $a = 0.019$ is a certified conditional rate; no numerical grid is needed for this result.

For comparison, define the normalized symmetric matrix $K = \frac{1}{2}M^{-1/2}FM^{-1/2}$. It is affine in (E,S,q), and $\lambda_{\max}(K)$ is convex. Its maximum over $[0,1]\times[0.001,1]\times[0,0.5]$ is therefore attained at a vertex. A floating-point check of all eight vertices gives $\max \lambda_{\max}(K) \approx -0.1292064$. This is a stronger numerical bound for the same fixed field, not an observed convergence rate or an equilibrium spectral abscissa. The exact 0.019 certificate above is the retained analytical guarantee. Reproduction code accompanies the source in `scripts/audits/contraction/`.

Equilibria and Invariance

For fixed deterministic forcing, any interior equilibrium of the unprojected model must satisfy

$$V^* = \frac{\kappa(E^*-I^*)}{\delta}, \tag{52}$$

$$E^* = \frac{\alpha I^* + \gamma_E\|\Delta\eta\|^2}{\alpha+\beta_E S^*}, \tag{53}$$

$$I^* = \frac{\beta_I C(V^*) - kS^*}{\gamma_I}, \tag{54}$$

$$S^* = \frac{\lambda_1\|\Delta\eta\|^2 - \lambda_2 C(V^*) + \beta_c\mathcal{C}}{\mu}. \tag{55}$$

These are necessary relations, not an existence proof. Under the certificate assumptions there can be at most one such equilibrium in $\mathcal{D}$: two distinct constant solutions would contradict the incremental bound. If one exists, other solutions remaining in $\mathcal{D}$ for all future time converge to it.

Existence and invariance cannot be inferred from positive damping. For example, zero drift and zero complexity give $\dot{S} = -\mu S - \lambda_2 C(V) < 0$ throughout $S \ge 0.001$ for the printed parameters. There is no equilibrium in $\mathcal{D}$ for that forcing, and the vector field points through its lower S boundary. A projected model has different equilibrium conditions. With time-varying inputs, contraction can hold without a fixed equilibrium at all.

C I-Channel Self-Regulation: Linear vs. Logistic

For this scalar comparison hold S and V fixed, and write $A = -kS + \beta_I C(V)$. The logistic alternative is $\dot{I} = A - \gamma_I I(1-I)$. If $0 < A < \gamma_I/4$, its two interior equilibria are

$$I^*_\pm = \tfrac{1}{2} \pm \tfrac{1}{2}\sqrt{1-4A/\gamma_I}. \tag{56}$$

The lower root is stable and the upper root unstable, since $\partial \dot{I}/\partial I = -\gamma_I(1-2I)$. This derivative is zero at $I = 0.5$ and equals $-\gamma_I$ and $+\gamma_I$ at $I = 0$ and $I = 1$, respectively.

Proposition C.1 (Sufficient Logistic Saturation Condition)**.** *For constant $A > \gamma_I/4$ and $I(0) \in [0,1)$, the unprojected scalar solution reaches $I = 1$ in finite time, since $\dot{I} \geq A - \gamma_I/4 > 0$ on $[0,1]$. Under continuous projection to $[0,1]$, it remains at that boundary afterward.*

The diagnostic

$$m_{\text{sat}} = 1 - 4A/\gamma_I \tag{57}$$

records this sufficient condition when negative. It is not a complete safety or boundary-distance test. At zero margin, a start below 0.5 approaches 0.5. At positive margin with $0 < A < \gamma_I/4$, a start above the unstable upper root still moves toward 1. For time-varying forcing, a snapshot of A does not establish persistence of a saturation condition.

The linear alternative $\dot{I} = A - \gamma_I I$ has equilibrium

$$I^* = A/\gamma_I, \tag{58}$$

which lies in $(0,1)$ exactly when $0 < A < \gamma_I$. Its derivative is the constant $-\gamma_I$. The reported linear parameter $\gamma_I = 0.169$ gives $I^* \approx 0.80$ when $A \approx 0.135$; this local tuning statement is not a uniform invariance guarantee.

Historical logistic diagnostic. The following retained snapshot evaluates the diagnostic for 75 active agents at $\gamma_I = 0.25$, $k = 0.10$, $\beta_I = 0.30$:

Regime	Mean m_{sat}	% negative
Convergence ($n = 24$)	-1.45	100%
Exploration ($n = 22$)	-1.18	100%
Divergence ($n = 29$)	-1.10	100%
All active ($n = 75$)	-1.23	100%

Every recorded forcing value exceeds $\gamma_I/4 = 0.0625$ under that parameterization. These diagnostic values are not observations of every agent reaching the boundary, and they do not establish behavior under other parameter histories.

D Boundary Clamping and Numerical Scope

The historical snapshot records 19 of 903 agents (2.1%) at $I = 1$, all in the `CONVERGENCE` regime with few updates. This is descriptive evidence about that snapshot, not a proof that the continuous field preserves its interior. The lower entropy floor is implemented by a numerical constraint; it cannot supply invariance to the unprojected equations.

Projection and finite steps. Coordinate clipping to a common box is nonexpansive in a constant positive diagonal metric. Thus projection after an already-contracting map preserves its contraction factor. It does not make an arbitrary Euler or Runge–Kutta step contracting. For example, $\dot{x} = -x$ contracts continuously, but Euler with $dt = 3$ followed by clipping to $[-1,1]$ doubles the distance between $x = 0.1$ and $x = 0$. A continuous projected system and a finite clipped integrator require distinct analyses.

Reproducible implementation counterexample. At UNITARES source revision `6d742273434c`, the default RK4 map with active linear-I parameters, zero drift and noise, complexity 0.5, no sensor coupling and zero entropy setpoint gives the following for $dt = 0.1$:

$$\begin{aligned} x &= (0.5, 0.999, 0.3, 0), \\ y &= (0.5, 0.9990001, 0.3, 0), \\ \|x - y\|_M &\approx 4.47214 \times 10^{-8}, \\ \|\operatorname{step}(x) - \operatorname{step}(y)\|_M &\approx 2.05954 \times 10^{-7}. \end{aligned}$$

The distance ratio is about 4.60527, rather than the proposed $e^{-0.019(0.1)} \approx 0.99810$. Both starts are interior; no stage/final clipping or entropy-floor activation occurs in this example. The soft barrier introduces strong local damping and numerical stiffness; removing only that barrier in the isolated counterfactual gives a ratio about 0.983613.

The reproducer in `scripts/audits/contraction/` exports the pinned modules from Git without reading

deployment data. This counterexample refutes universal contraction of that map; it does not establish frequency, reachability, asymptotic divergence or instability of the full governor. Later numerical fixes require separate validation.

Diagnostics and controller scope. No implemented threshold-tightening path tied to $m_{\mathrm{sat}} < 0.1$ was located; the earlier preventive claim is withdrawn. Certifying the update requires including barriers, clipping, feedback and adaptive parameters. Snapshot occupancy and saturation diagnostics do not replace that analysis.